36-671 Final Project

Ellie Huang, Nikita Jayaprakash, Annie Wang, Aishani Pradhan, Steve Wong

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Research Question

How do regional gasoline prices co-move over time, and what are the lead-lag relationships between regions and between crude oil prices and retail gasoline prices?

1. Preparing Time Series

```
library(here)
```

```
here() starts at /Users/nikita/Desktop/36671_Final_Project
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
library(readxl)
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.5.2


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.5.2
```

```r
library(ggplot2)
library(GGally)
library(vars)
```

```
Loading required package: MASS


Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select

Loading required package: strucchange

Loading required package: zoo


Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric

Loading required package: sandwich

Loading required package: urca

Loading required package: lmtest
```

```r
library(forecast)
```

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

```r
library(lubridate)
```

```
Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```r
file_path <- here("data", "PET_PRI_GND_A_EPM0_PTE_DPGAL_W.xls")
brent_path <- here("data", "brent_crude_weekly_1993_present.csv")
```

```r
# Load weekly gasoline data
# skip = 2 because row 1 = title, row 2 = source key, row 3 = actual variable names
us_data <- read_excel(file_path, sheet = "Data 1", skip = 2) |>
  rename(Date = 1, US_All_Grades = 2) |>
  mutate(Date = as.Date(Date), US_All_Grades = as.numeric(US_All_Grades)) |>
  arrange(Date)

region_data <- read_excel(file_path, sheet = "Data 2", skip = 2) |>
  rename(
    Date = 1,
    East_Coast = 2,
    New_England = 3,
    Central_Atlantic = 4,
    Lower_Atlantic = 5,
    Midwest = 6,
    Gulf_Coast = 7,
    Rocky_Mountain = 8,
    West_Coast = 9,
    West_Coast_ex_CA = 10
  ) |>
  mutate(Date = as.Date(Date)) |>
  mutate(across(-Date, as.numeric)) |>
```

```r
  arrange(Date)

# Join national + five regional series, drop any incomplete weeks
gas_complete <- us_data |>
  left_join(
    region_data |> dplyr::select(Date, East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_
    by = "Date"
  ) |>
  drop_na()

cat("Weekly gasoline date range:", min(gas_complete$Date), "to", max(gas_complete$Date), "\n"
```

```
Weekly gasoline date range: 8495 to 20542
```

```r
cat("Observations:", nrow(gas_complete), "\n")
```

```
Observations: 1722
```

```r
# Load weekly Brent crude oil data
crude_data <- read_csv(brent_path) |>
  rename(Brent = Brent_Price_USD) |>
  mutate(Date = as.Date(Date), Brent = as.numeric(Brent)) |>
  arrange(Date)
```

```
Rows: 1734 Columns: 2
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl  (1): Brent_Price_USD
date (1): Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
cat("Crude oil date range:", min(crude_data$Date), "to", max(crude_data$Date), "\n")
```

```
Crude oil date range: 8404 to 20535
```

```r
# Convert all series to weekly ts objects
# Get week number via %U (Sunday start); +1 to make it start from 1
start_vec <- c(
  as.numeric(format(min(gas_complete$Date), "%Y")),
  as.numeric(format(min(gas_complete$Date), "%U")) + 1
)

us_ts <- ts(gas_complete$US_All_Grades, start = start_vec, frequency = 52)
east_ts <- ts(gas_complete$East_Coast, start = start_vec, frequency = 52)
midwest_ts <- ts(gas_complete$Midwest, start = start_vec, frequency = 52)
gulf_ts <- ts(gas_complete$Gulf_Coast, start = start_vec, frequency = 52)
rocky_ts <- ts(gas_complete$Rocky_Mountain, start = start_vec, frequency = 52)
west_ts <- ts(gas_complete$West_Coast, start = start_vec, frequency = 52)

# Build multivariate ts for VAR analysis
regions_ts <- gas_complete |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  as.matrix() |>
  ts(start = start_vec, frequency = 52)

colnames(regions_ts) <- c("East_Coast", "Midwest", "Gulf_Coast", "Rocky_Mountain", "West_Coas
```

```r
# Both series are weekly, join directly on date
gas_crude <- gas_complete |>
  inner_join(crude_data, by = "Date") |>
  drop_na()

cat("Gas price-crude oil overlapping date range:", min(gas_crude$Date), "to", max(gas_crude$D
```

```
Gas price-crude oil overlapping date range: 8495 to 20535
```

```r
cat("Observations:", nrow(gas_crude), "\n")
```

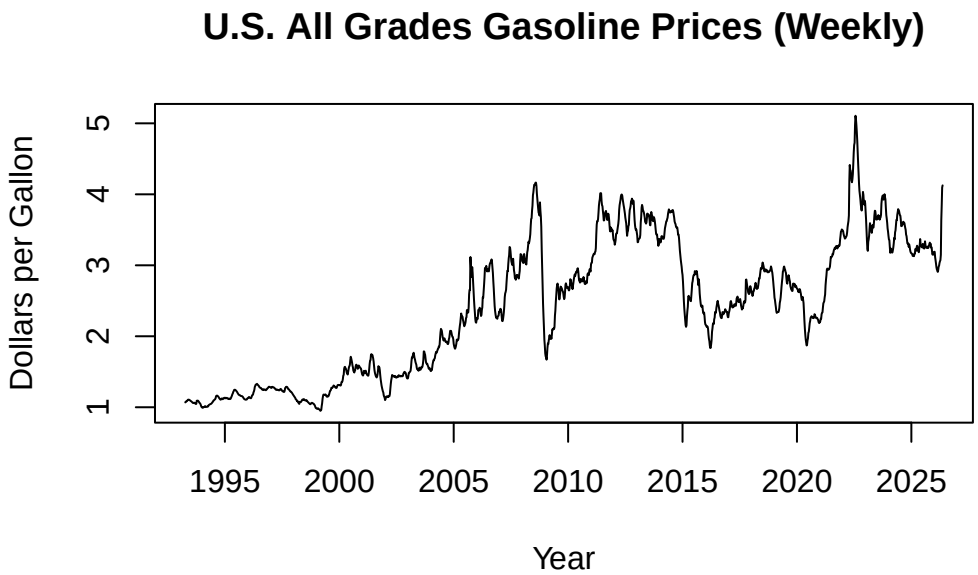
```
Observations: 1721
```

```r
# Shared start for weekly ts
start_crude <- c(
  as.numeric(format(min(gas_crude$Date), "%Y")),
  as.numeric(format(min(gas_crude$Date), "%U")) + 1
)
```

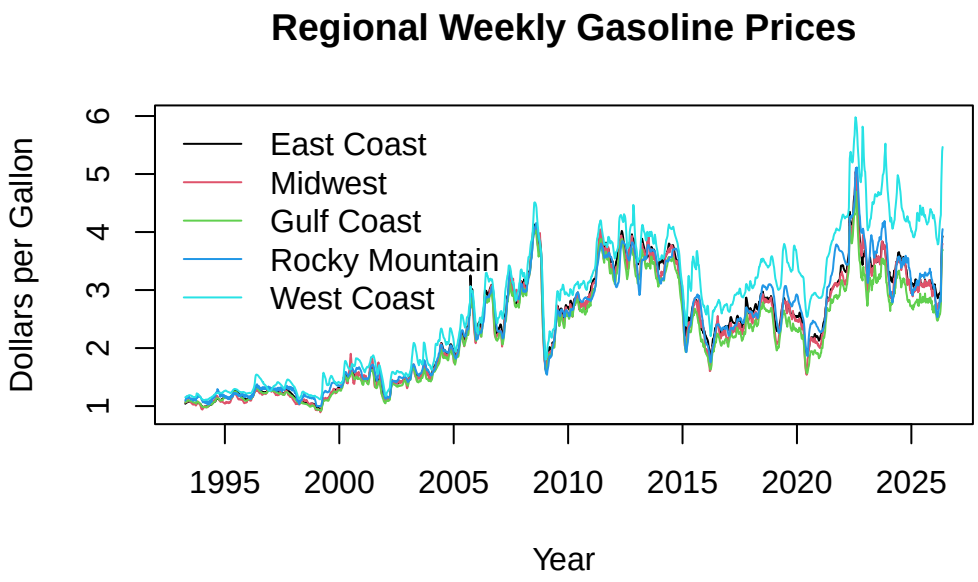
```r
gas_ts_crude <- ts(gas_crude$US_All_Grades, start = start_crude, frequency = 52)
crude_ts <- ts(gas_crude$Brent, start = start_crude, frequency = 52)
```

2. Exploratory Data Analysis

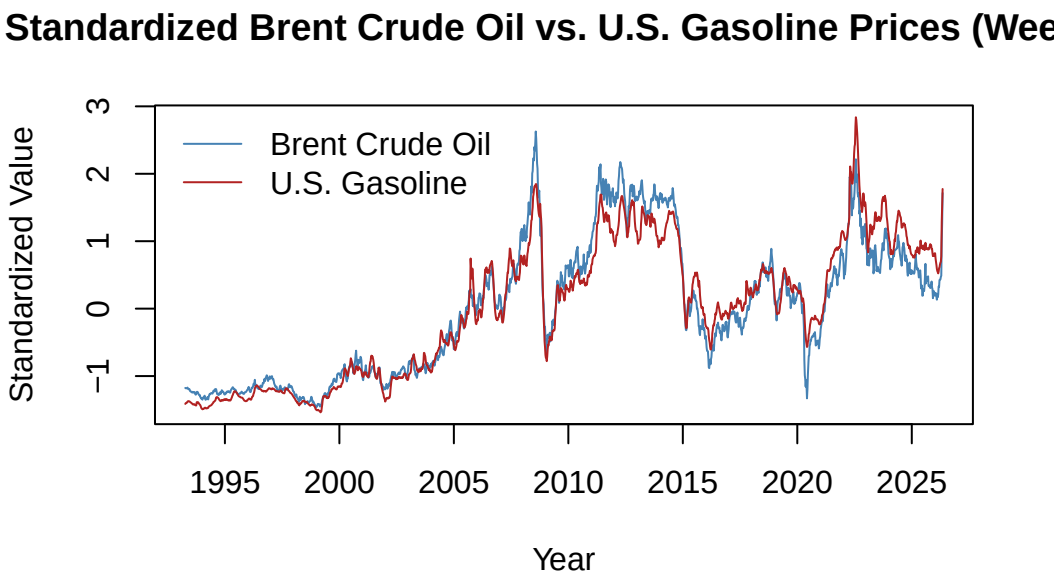
```r
# Time series plots: national, regional, and crude oil
plot(us_ts,
     main = "U.S. All Grades Gasoline Prices (Weekly)",
     xlab = "Year", ylab = "Dollars per Gallon")
```




```r
ts.plot(east_ts, midwest_ts, gulf_ts, rocky_ts, west_ts,
        col = 1:5, lty = 1,
        main = "Regional Weekly Gasoline Prices",
        xlab = "Year", ylab = "Dollars per Gallon")
legend("topleft",
       legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
       col = 1:5, lty = 1, bty = "n")
```




```r
# Standardize for comparison since crude oil and gas prices are on diff scales/units
crude_scaled <- ts(
  as.numeric(scale(as.numeric(crude_ts))),
  start = start(crude_ts),
  frequency = frequency(crude_ts)
)

gas_scaled <- ts(
  as.numeric(scale(as.numeric(gas_ts_crude))),
  start = start(gas_ts_crude),
  frequency = frequency(gas_ts_crude)
)

ts.plot(crude_scaled, gas_scaled,
        col = c("steelblue", "firebrick"), lty = 1,
        main = "Standardized Brent Crude Oil vs. U.S. Gasoline Prices (Weekly)",
        xlab = "Year", ylab = "Standardized Value")
legend("topleft",
       legend = c("Brent Crude Oil", "U.S. Gasoline"),
       col = c("steelblue", "firebrick"), lty = 1, bty = "n")
```



The national gasoline series shows long swings over time rather than fluctuating around a constant level, suggesting strong persistence and possible nonstationarity in the level series. There seems to be a general upward movement over the sample. The plots also suggest changing variability over time, with larger swings during some periods, especially around major shocks, which may indicate heteroskedasticity.

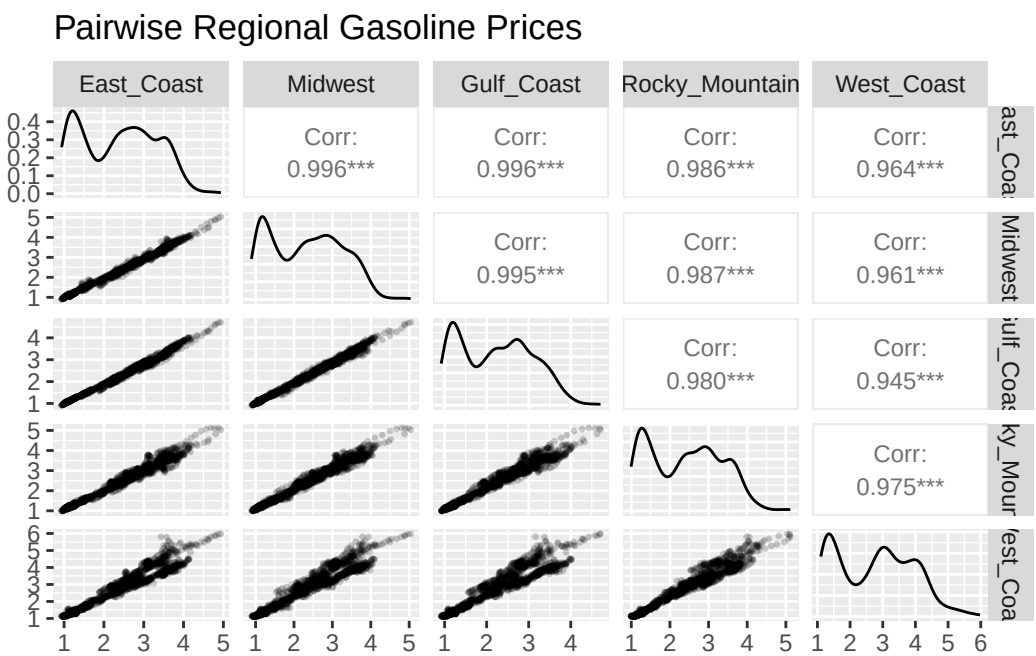
The regional plot shows that all five regions follow the same broad pattern, rising and falling during the same major episodes.

At the same time, the regions are not exactly the same. The West Coast is generally higher than the other regions and appears more volatile, especially in later years. Rocky Mountain also shows somewhat greater variability at certain points, while the Gulf Coast and Midwest tend to track the national pattern more closely and at lower price levels. So the data show both strong shared movement and meaningful regional differences.

Comparing Brent crude to U.S. gasoline, the two series move very closely together over time. This suggests that crude oil is an important driver of gasoline prices, although there are other factors like refining, transportation, taxes, regional conditions, etc.

Question: Ask prof if we need to include these scatterplots too, or just the lagged scatters.

```r
# Pairwise scatterplots to check contemporaneous correlations before looking at lags
gas_complete |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  ggpairs(
    title = "Pairwise Regional Gasoline Prices",
    lower = list(continuous = wrap("points", alpha = 0.2, size = 0.5)),
    upper = list(continuous = wrap("cor", size = 3))
  )
```



The pairwise scatterplots indicate strong positive contemporaneous relationships among weekly gasoline prices across all regions. For every regional pair, the points form a clear upward-sloping pattern, suggesting that when prices are relatively high in one region, they also tend to be relatively high in the other during the same week. This is consistent with the very high pairwise correlations, which range from 0.945 to 0.996. Several pairs, such as East Coast and Midwest, fall especially tightly along a line, indicating very strong same-week linear association. Pairs involving the West Coast appear somewhat more dispersed by comparison, though the relationship remains strongly positive. However, because these plots compare observations from the same week only, they do not provide evidence about lead-lag structure or whether price changes in one region tend to precede changes in another.

This shows that the regions are clearly not independent; therefore it is reasonable to study them as a connected multivariate time series. Having established strong contemporaneous correlation, we can next investigate whether some regions tend to lead or lag others over

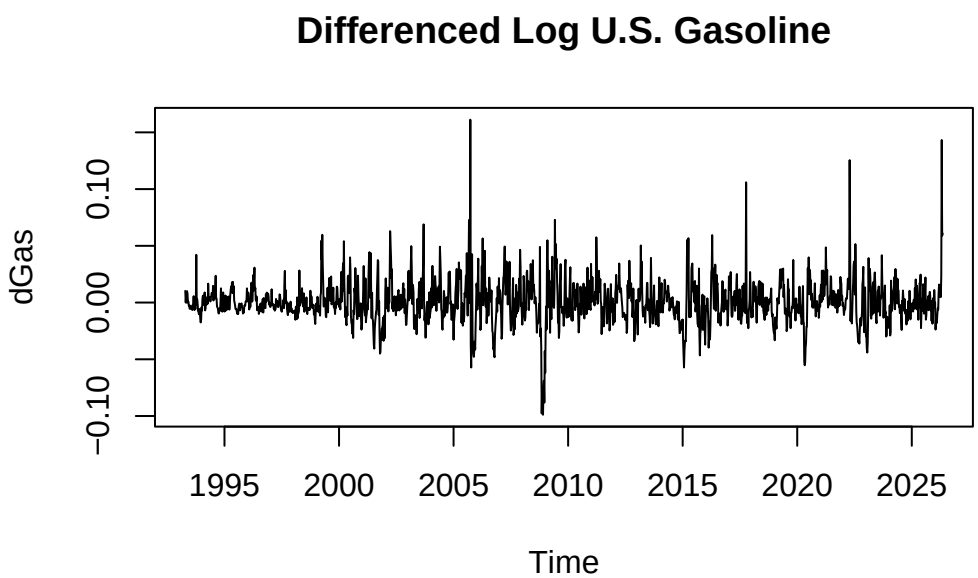
time. This provides a natural motivation for using lag-based analysis and vector autoregressive models.

3. Crude Oil Prices vs US Gasoline Price Analyses

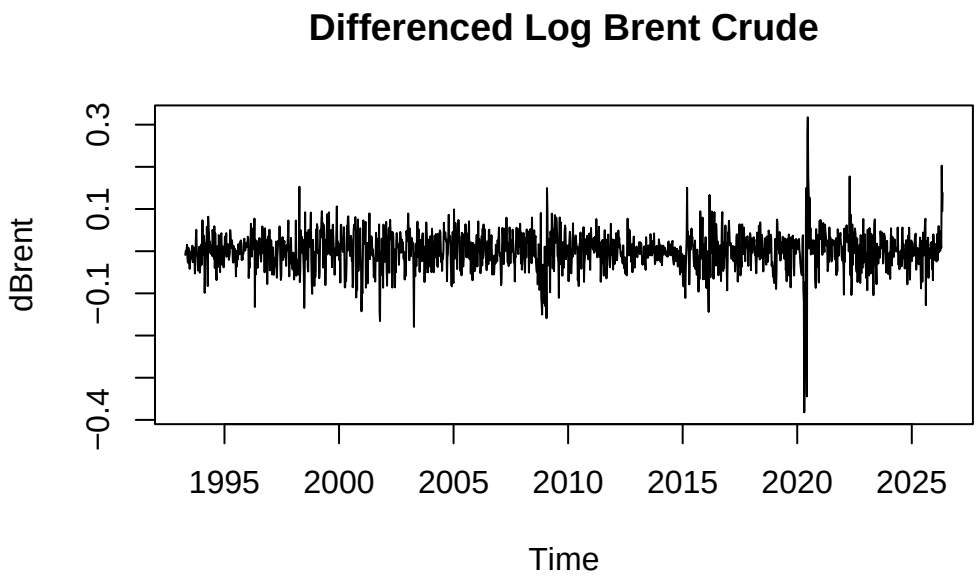
(i) Converting to Stationary Series

Both crude oil and gasoline prices exhibit strong trends over time, so we first transform the series to make them approximately stationary before analyzing their lag relationships.

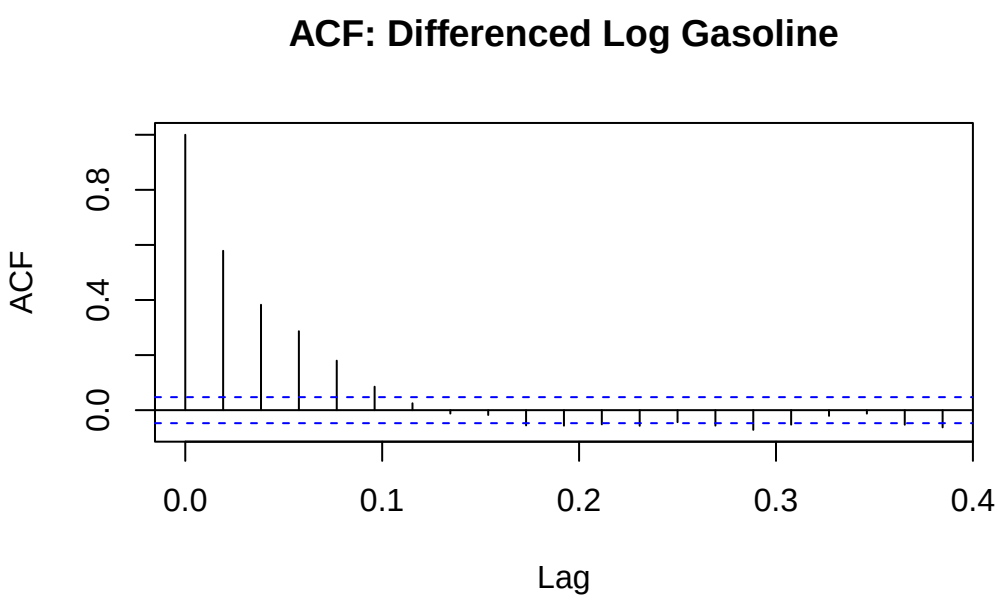
```
# Log difference
dGas <- diff(log(gas_ts_crude))
dBrent <- diff(log(crude_ts))

#par(mfrow = c(3,2))
plot(dGas, main = "Differenced Log U.S. Gasoline")
```



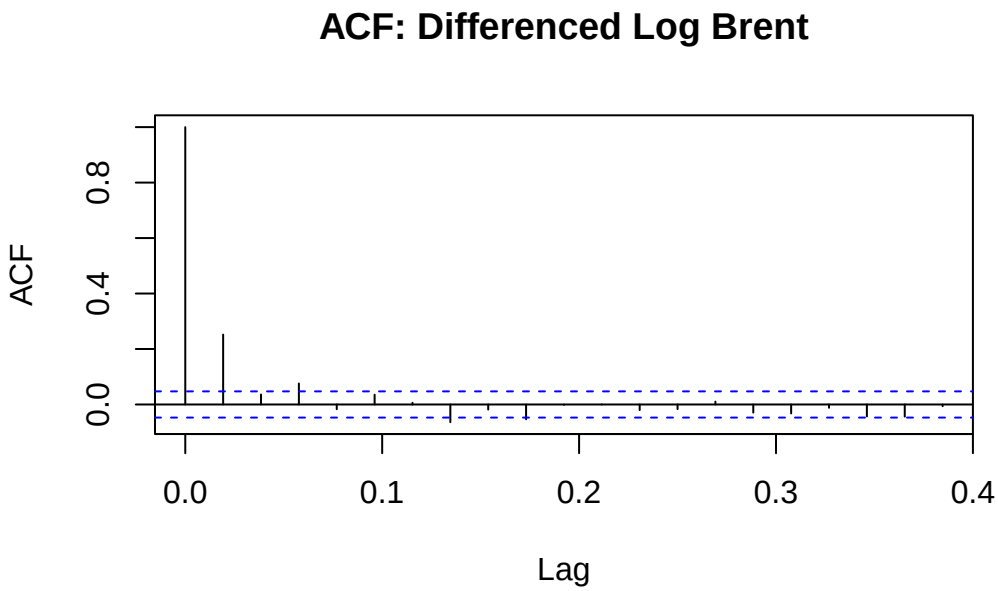

```
plot(dBrent, main = "Differenced Log Brent Crude")
```



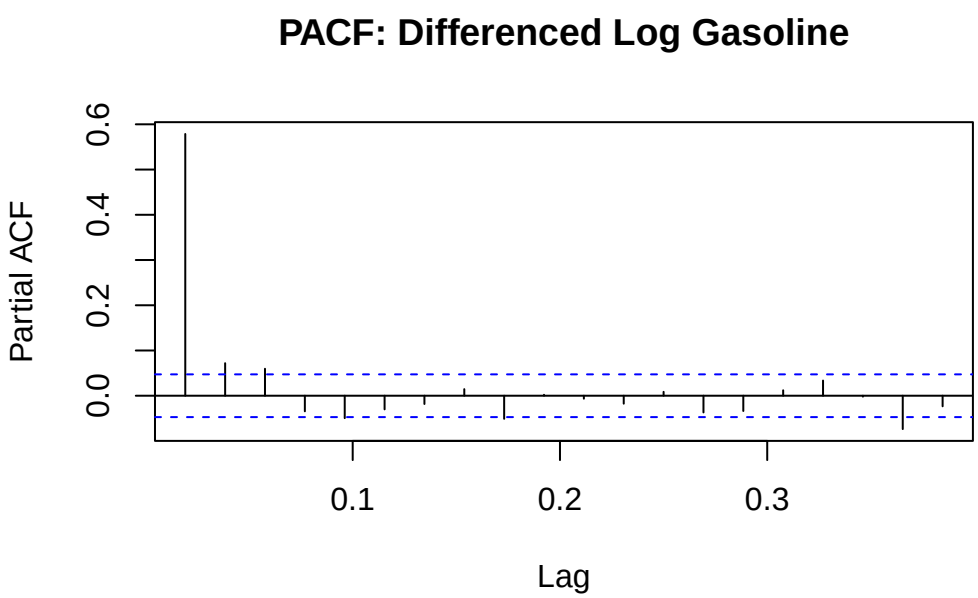

```r
acf(dGas, lag.max = 20, main = "ACF: Differenced Log Gasoline")
```



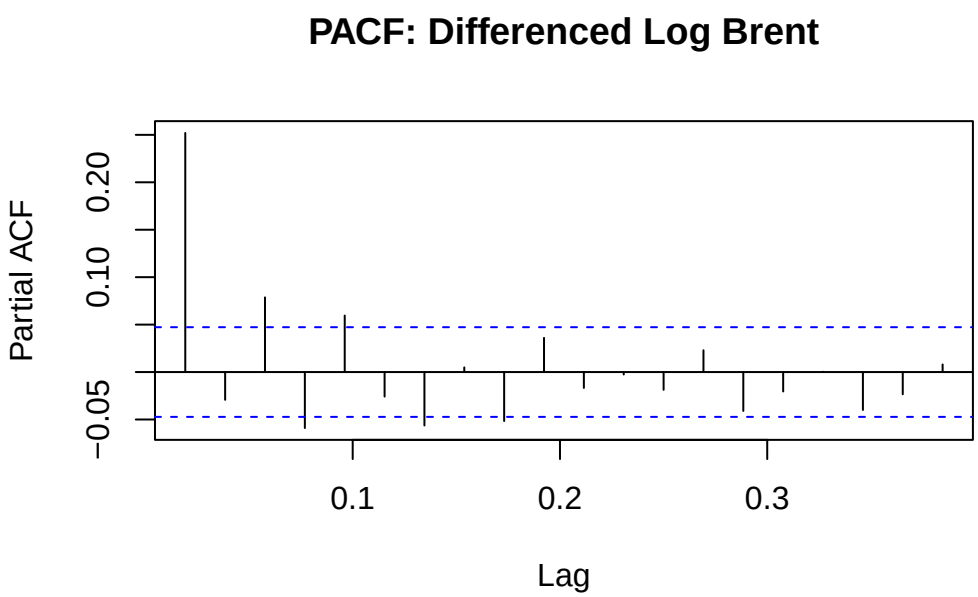
```r
acf(dBrent, lag.max = 20, main = "ACF: Differenced Log Brent")
```




```r
pacf(dGas, lag.max = 20, main = "PACF: Differenced Log Gasoline")
```



```r
pacf(dBrent, lag.max = 20, main = "PACF: Differenced Log Brent")
```

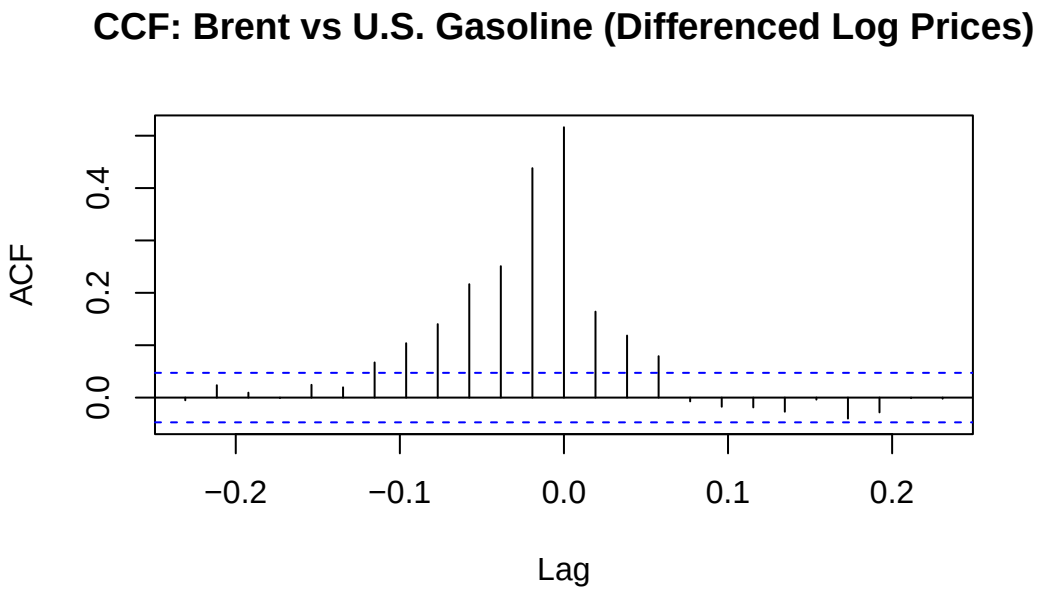



```r
par(mfrow = c(1,1))
```

After taking first differences of the log prices, both series fluctuate around a stable mean with rapidly decaying ACFs. The PACFs show one dominant spike at lag 1 followed by little remaining dependence, which is consistent with a low order AR process and suggests that the transformed series are sufficiently stationary for CCF and VAR analysis.

(ii) CCF and LR to Find Lag Dependence

```r
# Negative lag means Brent leads gasoline
ccf(dBrent, dGas,
    lag.max = 12,
    main = "CCF: Brent vs U.S. Gasoline (Differenced Log Prices)")
```

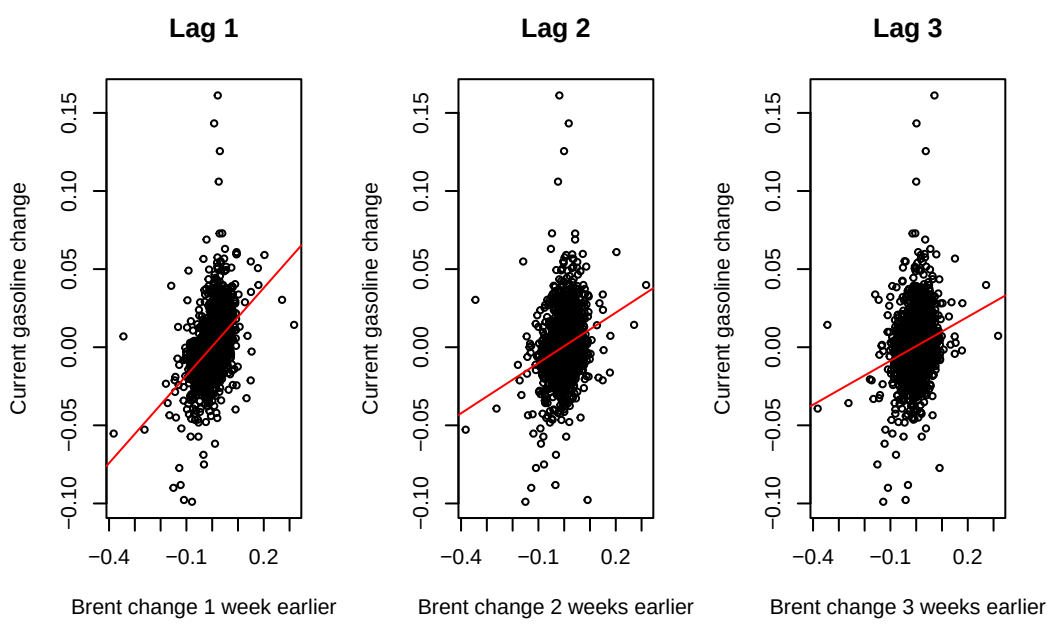


The cross-correlation of the differenced log prices is largest at lag 0, indicating that gasoline and crude oil prices move together strongly within the same week. There are also significant positive correlations at lags of approximately -1 to -3 weeks, suggesting that changes in Brent crude prices continue to pass through into gasoline prices over the following several weeks.

Question: how does the instantaneous correlation impact our analysis?

```
oil_gas_lag <- ts.intersect(
  gas = dGas,
  brentL1 = stats::lag(dBrent, -1),
  brentL2 = stats::lag(dBrent, -2),
  brentL3 = stats::lag(dBrent, -3),
  dframe = TRUE
)

# Scatterplots

par(mfrow = c(1,3))

plot(oil_gas_lag$brentL1, oil_gas_lag$gas,
     xlab = "Brent change 1 week earlier",
     ylab = "Current gasoline change",
     main = "Lag 1")
```

```r
abline(lm(gas ~ brentL1, data = oil_gas_lag), col = "red")

plot(oil_gas_lag$brentL2, oil_gas_lag$gas,
     xlab = "Brent change 2 weeks earlier",
     ylab = "Current gasoline change",
     main = "Lag 2")
abline(lm(gas ~ brentL2, data = oil_gas_lag), col = "red")

plot(oil_gas_lag$brentL3, oil_gas_lag$gas,
     xlab = "Brent change 3 weeks earlier",
     ylab = "Current gasoline change",
     main = "Lag 3")
abline(lm(gas ~ brentL3, data = oil_gas_lag), col = "red")
```




```r
par(mfrow = c(1,1))
```

```r
# Check if lagged valued of crude oil week 1-3 are signficant in predicting gas prices
fit_oil <- lm(gas ~ brentL1 + brentL2 + brentL3,
              data = oil_gas_lag)

summary(fit_oil)
```

```
Call:
lm(formula = gas ~ brentL1 + brentL2 + brentL3, data = oil_gas_lag)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.080184 -0.008914 -0.001070  0.007719  0.152296 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.0005041  0.0004021   1.254     0.21    
brentL1     0.1735182  0.0092325  18.794  < 2e-16 ***
brentL2     0.0449195  0.0095342   4.711 2.66e-06 ***
brentL3     0.0776113  0.0092899   8.354  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01665 on 1713 degrees of freedom
Multiple R-squared:  0.2452,	Adjusted R-squared:  0.2439 
F-statistic: 185.5 on 3 and 1713 DF,  p-value: < 2.2e-16
```

The lagged scatterplots and regression reinforce the CCF results. The relationship between crude oil and gasoline changes is strongest after one week, with weaker but still positive associations after two and three weeks. This supports the conclusion that gasoline prices adjust gradually over several weeks following a change in crude oil prices.

(iii) VAR(2) Model

```r
library(vars)

# Combine the stationary series
x_oil <- cbind(
  Gasoline = dGas,
  Brent = dBrent
)

# choose lag length
VARselect(x_oil, lag.max = 8, type = "none")
```

```
$selection
```

```
AIC(n)  HQ(n)  SC(n) FPE(n)
     5      3      2      5

$criteria
                   1             2             3             4             5
AIC(n) -1.491150e+01 -1.492793e+01 -1.493507e+01 -1.493885e+01 -1.494229e+01
HQ(n)  -1.490679e+01 -1.491851e+01 -1.492094e+01 -1.492001e+01 -1.491875e+01
SC(n)  -1.489878e+01 -1.490248e+01 -1.489690e+01 -1.488796e+01 -1.487867e+01
FPE(n)  3.342081e-07  3.287641e-07  3.264252e-07  3.251927e-07  3.240755e-07
                   6             7             8
AIC(n) -1.493893e+01 -1.493895e+01 -1.493744e+01
HQ(n)  -1.491068e+01 -1.490599e+01 -1.489977e+01
SC(n)  -1.486260e+01 -1.484989e+01 -1.483566e+01
FPE(n)  3.251649e-07  3.251606e-07  3.256512e-07
```

```r
fit_var_oil <- VAR(x_oil, p = 2, type = "none")
summary(fit_var_oil)
```

```
VAR Estimation Results:
=========================
Endogenous variables: Gasoline, Brent
Deterministic variables: none
Sample size: 1718
Log Likelihood: 7960.895
Roots of the characteristic polynomial:
0.6704 0.2709 0.2709 0.1272
Call:
VAR(y = x_oil, p = 2, type = "none")


Estimation results for equation Gasoline:
=========================================
Gasoline = Gasoline.l1 + Brent.l1 + Gasoline.l2 + Brent.l2

             Estimate Std. Error t value Pr(>|t|)
Gasoline.l1  0.411722   0.027724  14.851  < 2e-16 ***
Brent.l1     0.093528   0.009721   9.622  < 2e-16 ***
Gasoline.l2  0.141533   0.025792   5.488 4.69e-08 ***
Brent.l2    -0.023122   0.009733  -2.375   0.0176 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.01515 on 1714 degrees of freedom
Multiple R-Squared: 0.3757, Adjusted R-squared: 0.3742
F-statistic: 257.8 on 4 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Brent:
======================================
Brent = Gasoline.l1 + Brent.l1 + Gasoline.l2 + Brent.l2

             Estimate Std. Error t value Pr(>|t|)
Gasoline.l1  0.006147   0.079397   0.077 0.938301
Brent.l1     0.255444   0.027838   9.176  < 2e-16 ***
Gasoline.l2  0.294836   0.073864   3.992 6.84e-05 ***
Brent.l2    -0.092385   0.027875  -3.314 0.000938 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.0434 on 1714 degrees of freedom
Multiple R-Squared: 0.07644,    Adjusted R-squared: 0.07429
F-statistic: 35.47 on 4 and 1714 DF,  p-value: < 2.2e-16



Covariance matrix of residuals:
          Gasoline     Brent
Gasoline 0.0002296 0.0003275
Brent    0.0003275 0.0018832

Correlation matrix of residuals:
         Gasoline  Brent
Gasoline   1.0000 0.4981
Brent      0.4981 1.0000
```

Questions:

- Which VAR model to use in terms of AIC, HQ, SC, FPE? SC with parameter = 2 because simplest?

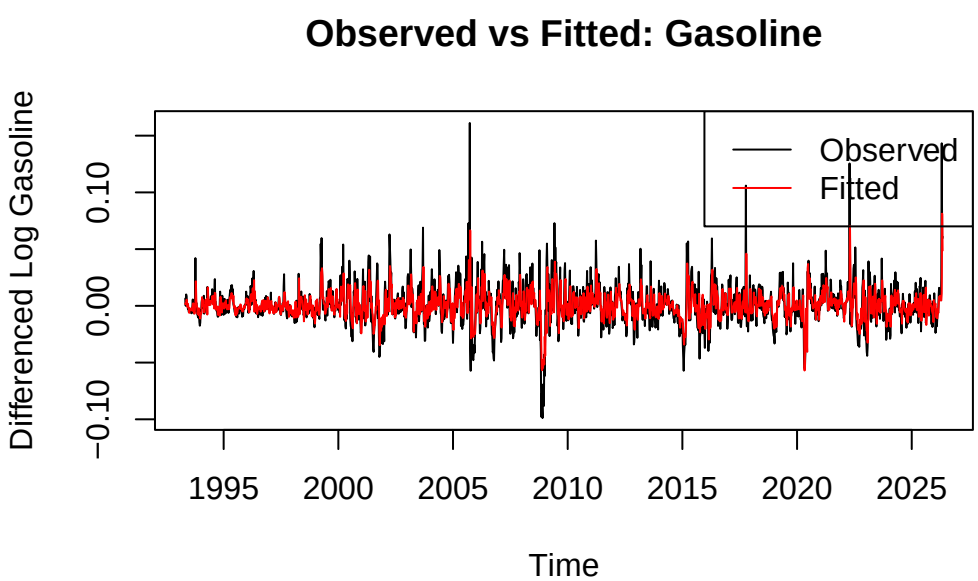
  `AIC(n)  HQ(n)  SC(n) FPE(n)`

- why would the effect of gasoline (l2) be stronger on crude oil than crude oil on gasoline (estimates)? remaining correlation? are the estimates even comparable?

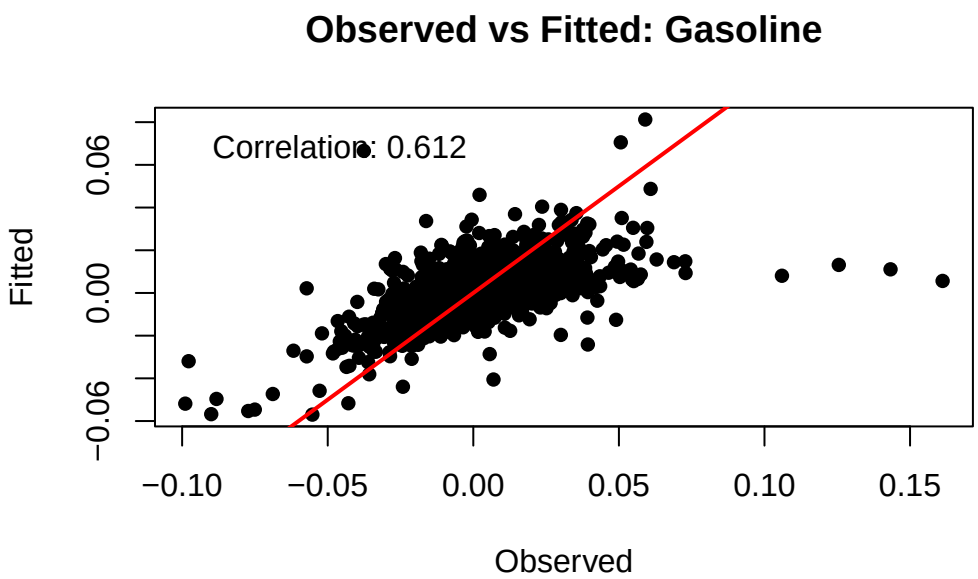
(iv) Diagnostics

```
# Fitted values and residuals from VAR(2)
fitted_vals <- fitted(fit_var_oil)
res <- resid(fit_var_oil)
```

```
# Observed and fitted gasoline changes
obs_gas <- window(x_oil[, "Gasoline"], start = time(x_oil)[3])
fit_gas <- ts(fitted_vals[, "Gasoline"],
              start = start(obs_gas),
              frequency = frequency(obs_gas))

plot(obs_gas, type = "l", col = "black",
     main = "Observed vs Fitted: Gasoline",
     ylab = "Differenced Log Gasoline", xlab = "Time")
lines(fit_gas, col = "red")
legend("topright",
       legend = c("Observed", "Fitted"),
       col = c("black", "red"), lwd = 1)
```



```r
plot(obs_gas, fit_gas,
     pch = 16,
     xlab = "Observed",
     ylab = "Fitted",
     main = "Observed vs Fitted: Gasoline")
legend("topleft",
       legend = paste("Correlation:",
                      round(cor(obs_gas, fit_gas), 3)),
       bty = "n")
abline(0, 1, col = "red", lwd = 2)
```




```r
## GAS
# top 10 residuals
res_gas <- res[,1]
time <- time(dGas)[-c(1,2)]
top_resid_gas <- order(abs(res_gas), decreasing = TRUE)[1:10]
# corresponding times in date format
top_resid_gas_time <- time[top_resid_gas] |>
  date_decimal() |>
  as.Date()
# in tabular format
tibble::tibble(
```

```r
  `Time (Gas)` = top_resid_gas_time,
  `Top Residuals (Gas)` = res_gas[top_resid_gas])
```

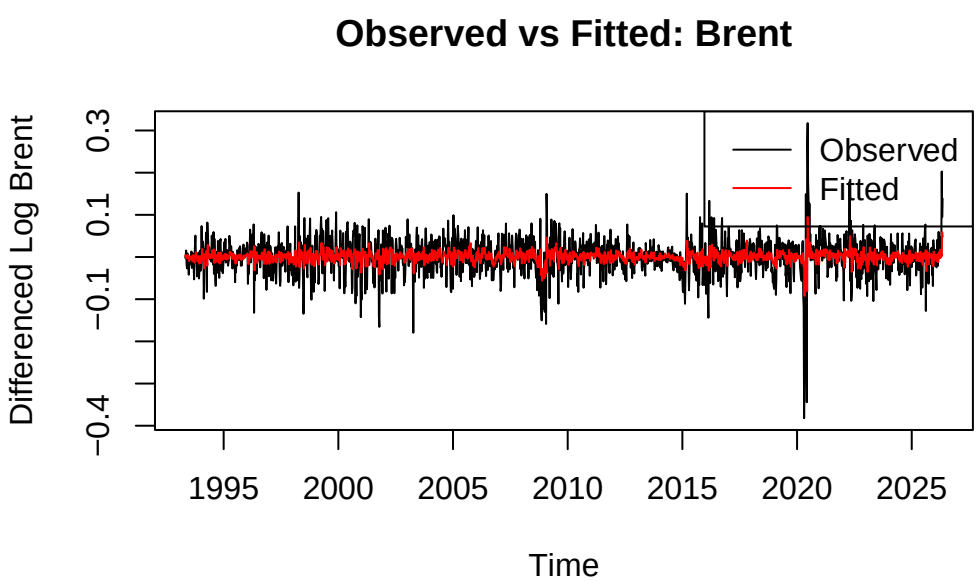
```
# A tibble: 10 x 2
   `Time (Gas)` `Top Residuals (Gas)`
   <date>                       <dbl>
 1 2005-09-24                  0.156
 2 2026-04-23                  0.132
 3 2022-04-16                  0.112
 4 2005-10-01                 -0.104
 5 2017-10-08                  0.0979
 6 2008-11-05                 -0.0658
 7 2009-06-04                  0.0636
 8 2009-01-29                  0.0635
 9 2008-10-08                  0.0616
10 2005-10-08                 -0.0594
```

```r
## BRENT
# top 10 residuals
res_brent <- res[,2]
time <- time(dBrent)[-c(1,2)]
top_resid_brent <- order(abs(res_brent), decreasing = TRUE)[1:10]
# corresponding times in date format
top_resid_brent_time <- time[top_resid_brent] |>
  date_decimal() |>
  as.Date()
# in tabular format
tibble::tibble(
  `Time (Brent)` = top_resid_brent_time,
  `Top Residuals (Brent)` = res_brent[top_resid_brent]
)
```

```
# A tibble: 10 x 2
   `Time (Brent)` `Top Residuals (Brent)`
   <date>                           <dbl>
 1 2020-04-22                      -0.355
 2 2020-06-10                       0.352
 3 2020-06-03                      -0.283
 4 2020-06-17                       0.223
 5 2020-05-20                       0.202
 6 2026-04-23                       0.201
```

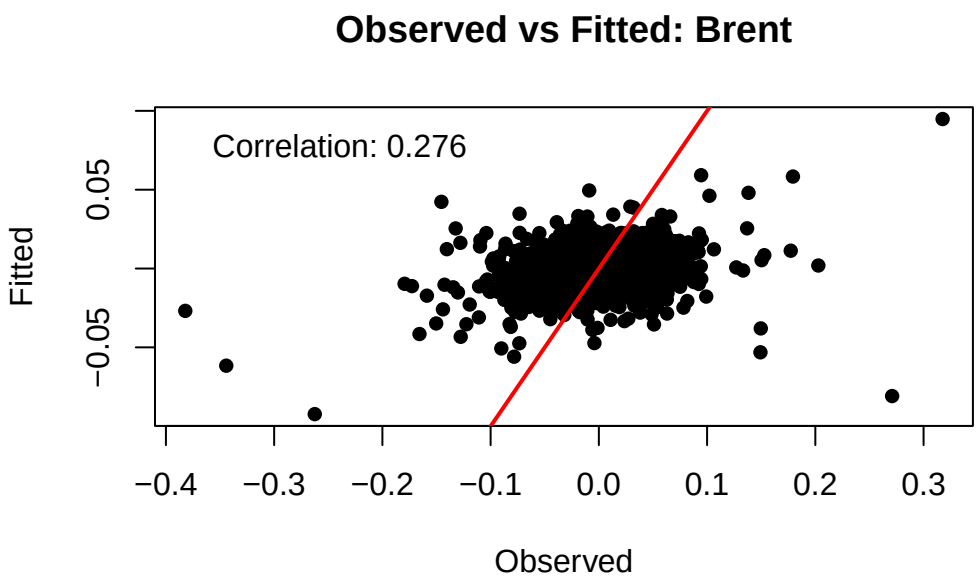
```
 7 2020-05-27                        -0.188
 8 2009-01-29                         0.188
 9 2020-04-29                        -0.170
10 2003-04-09                        -0.170
```

```r
# observed and fitted Brent changes
obs_brent <- window(x_oil[, "Brent"], start = time(x_oil)[3])
fit_brent <- ts(fitted_vals[, "Brent"],
                start = start(obs_brent),
                frequency = frequency(obs_brent))

plot(obs_brent, type = "l", col = "black",
     main = "Observed vs Fitted: Brent",
     ylab = "Differenced Log Brent", xlab = "Time")
lines(fit_brent, col = "red")
legend("topright",
       legend = c("Observed", "Fitted"),
       col = c("black", "red"), lwd = 1)
```



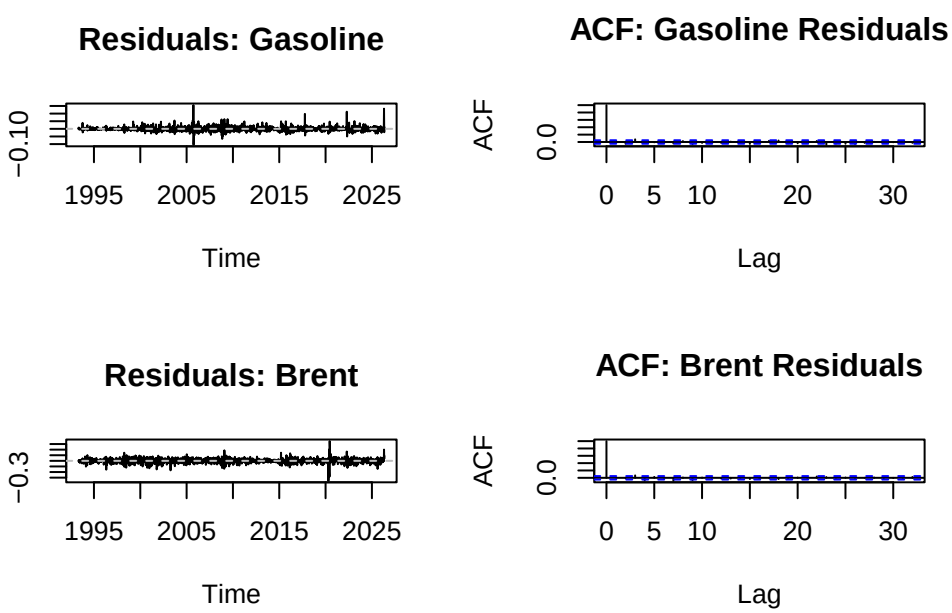

```r
plot(obs_brent, fit_brent,
     pch = 16,
     xlab = "Observed",
```

```r
     ylab = "Fitted",
     main = "Observed vs Fitted: Brent")
legend("topleft",
       legend = paste("Correlation:",
                      round(cor(obs_brent, fit_brent), 3)),
       bty = "n")
abline(0, 1, col = "red", lwd = 2)
```

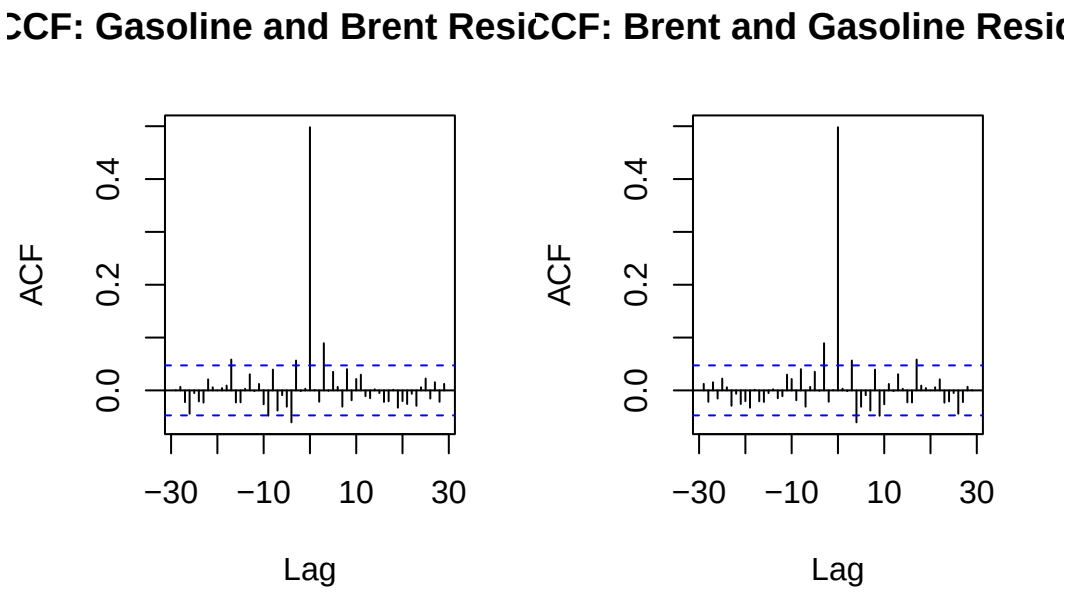



```r
# residual series
res_gas <- ts(res[, "Gasoline"],
              start = time(x_oil)[3],
              frequency = frequency(x_oil))

res_brent <- ts(res[, "Brent"],
                start = time(x_oil)[3],
                frequency = frequency(x_oil))

par(mfrow = c(2,2))

plot(res_gas, type = "l",
     main = "Residuals: Gasoline",
     ylab = "", xlab = "Time")
```

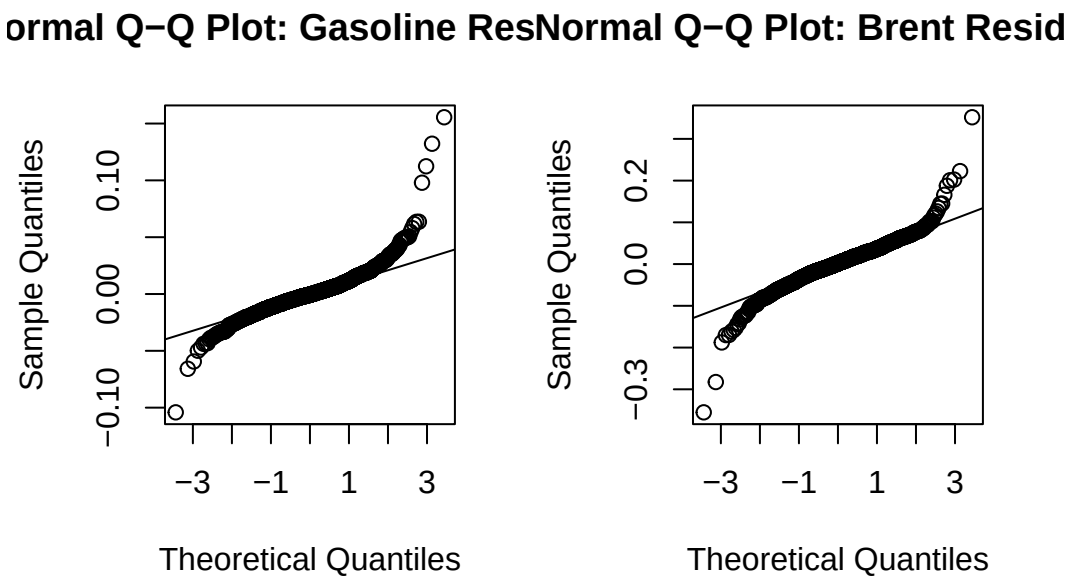
```r
abline(h = 0, lty = 2, col = "gray")

acf(res[, "Gasoline"],
    main = "ACF: Gasoline Residuals")

plot(res_brent, type = "l",
     main = "Residuals: Brent",
     ylab = "", xlab = "Time")
abline(h = 0, lty = 2, col = "gray")

acf(res[, "Brent"],
    main = "ACF: Brent Residuals")
```




```r
# residual cross-correlation
par(mfrow = c(1,2))

ccf(res[, "Gasoline"], res[, "Brent"],
    main = "CCF: Gasoline and Brent Residuals")

ccf(res[, "Brent"], res[, "Gasoline"],
    main = "CCF: Brent and Gasoline Residuals")
```




```r
# normal Q-Q plots
par(mfrow = c(1,2))

qqnorm(res[, "Gasoline"],
       main = "Normal Q-Q Plot: Gasoline Residuals")
qqline(res[, "Gasoline"])

qqnorm(res[, "Brent"],
       main = "Normal Q-Q Plot: Brent Residuals")
qqline(res[, "Brent"])
```




```
par(mfrow = c(1,1))
```

Overall, the diagnostic plots suggest that the VAR(2) model provides an adequate fit to the differenced gasoline and crude oil series. The fitted values track the observed series reasonably well, particularly for gasoline, and the residuals fluctuate around zero without obvious trends or structure. The residual ACFs and CCFs indicate little remaining serial or cross-correlation beyond lag 0, suggesting that the model captures the main lead–lag relationships. Although the Q-Q plots show some deviations in the tails, especially for Brent, the residuals are otherwise reasonably well behaved, so there is no major evidence of model misspecification.

Questions: This doesn't seem like the best model for brent with gas as covariate, but is that relevant to us? since we care more about brent predicting gas.

(v) Granger Causality to Further Investigate Direction

```
causality(fit_var_oil, cause = "Brent")
```

```
$Granger
```

```
    Granger causality H0: Brent do not Granger-cause Gasoline

data:  VAR object fit_var_oil
F-Test = 47.151, df1 = 2, df2 = 3428, p-value < 2.2e-16


$Instant

    H0: No instantaneous causality between: Brent and Gasoline

data:  VAR object fit_var_oil
Chi-squared = 341.64, df = 1, p-value < 2.2e-16
```

```r
causality(fit_var_oil, cause = "Gasoline")
```

```
$Granger

    Granger causality H0: Gasoline do not Granger-cause Brent

data:  VAR object fit_var_oil
F-Test = 10.806, df1 = 2, df2 = 3428, p-value = 2.097e-05


$Instant

    H0: No instantaneous causality between: Gasoline and Brent

data:  VAR object fit_var_oil
Chi-squared = 341.64, df = 1, p-value < 2.2e-16
```

The Granger causality tests indicate bidirectional predictive relationships between crude oil and gasoline prices. However, the predictive effect from oil to gasoline is substantially stronger than the reverse. This suggests that crude oil prices are the primary driver of gasoline price changes, while the predictive power of gasoline for oil likely reflects common shocks and market comovement rather than a strong causal effect.

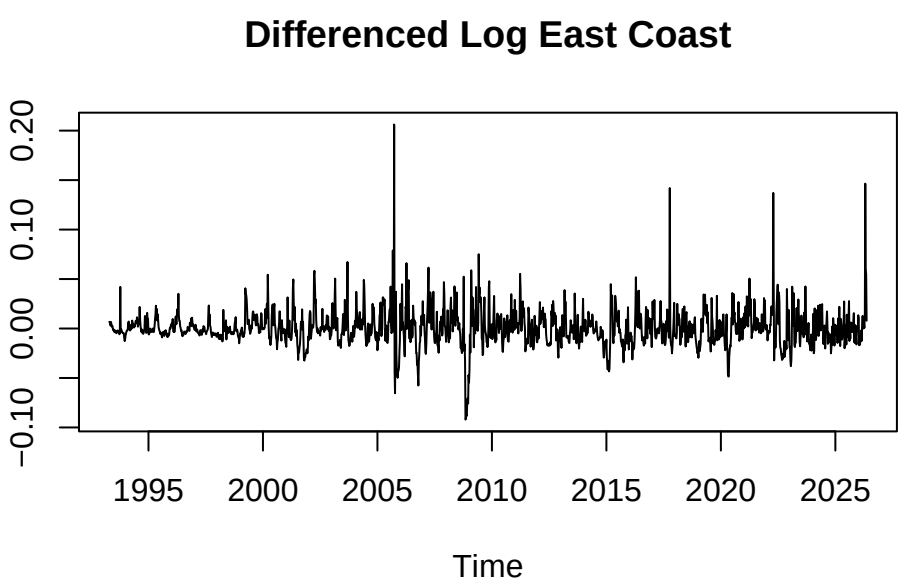
(vi) Forecast?

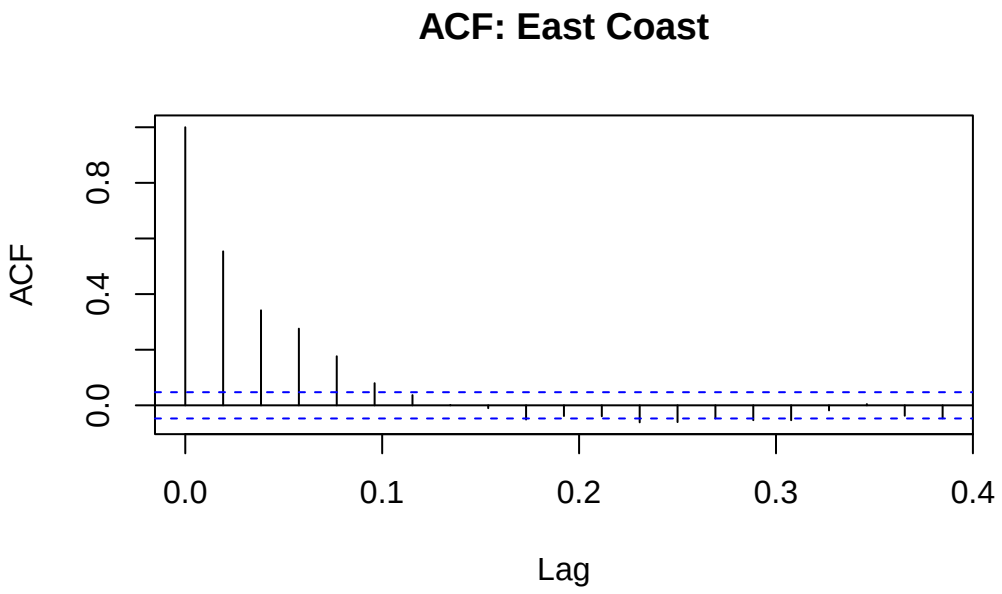
Question: how long do we want to forecast?

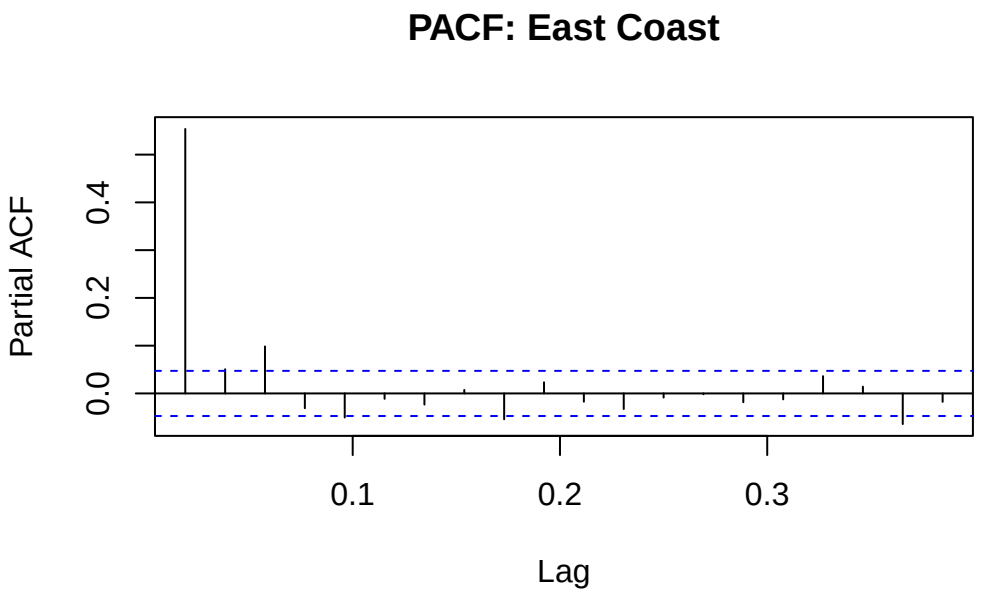
4. US Gasoline Prices Regional Comparison

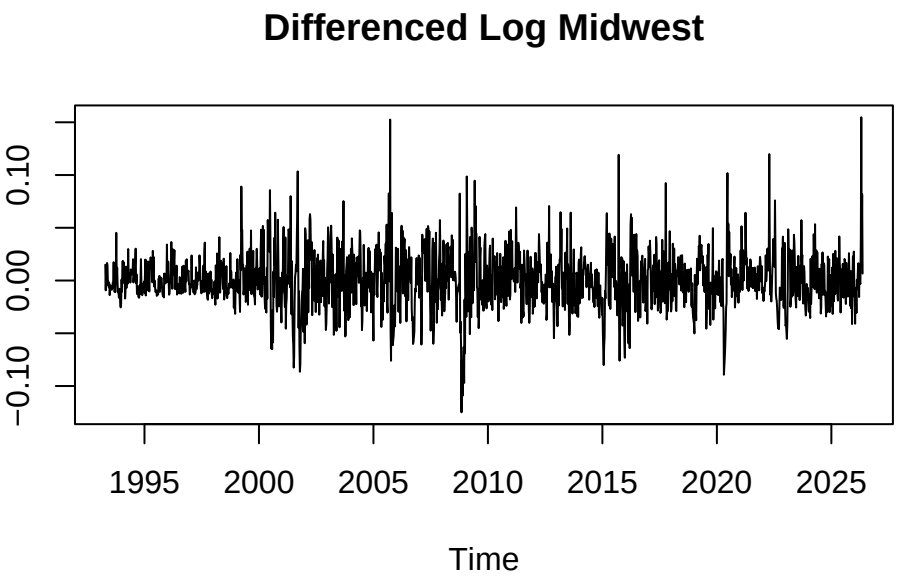
(i) Converting to Stationary Series

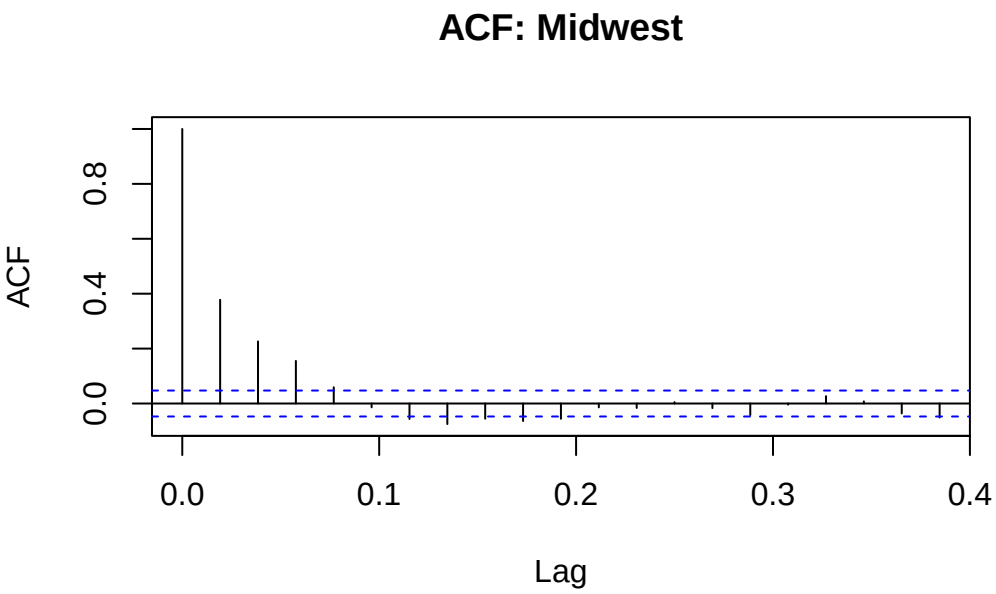
```
# First-difference the log prices for all 5 regions
dRegions <- diff(log(regions_ts))
colnames(dRegions) <- colnames(regions_ts)

for (nm in colnames(dRegions)) {
  plot(dRegions[, nm],
       main = paste("Differenced Log", gsub("_", " ", nm)),
       ylab = "", xlab = "Time")

  acf(dRegions[, nm],
      lag.max = 20,
      main = paste("ACF:", gsub("_", " ", nm)))

  pacf(dRegions[, nm],
       lag.max = 20,
       main = paste("PACF:", gsub("_", " ", nm)))
}
```

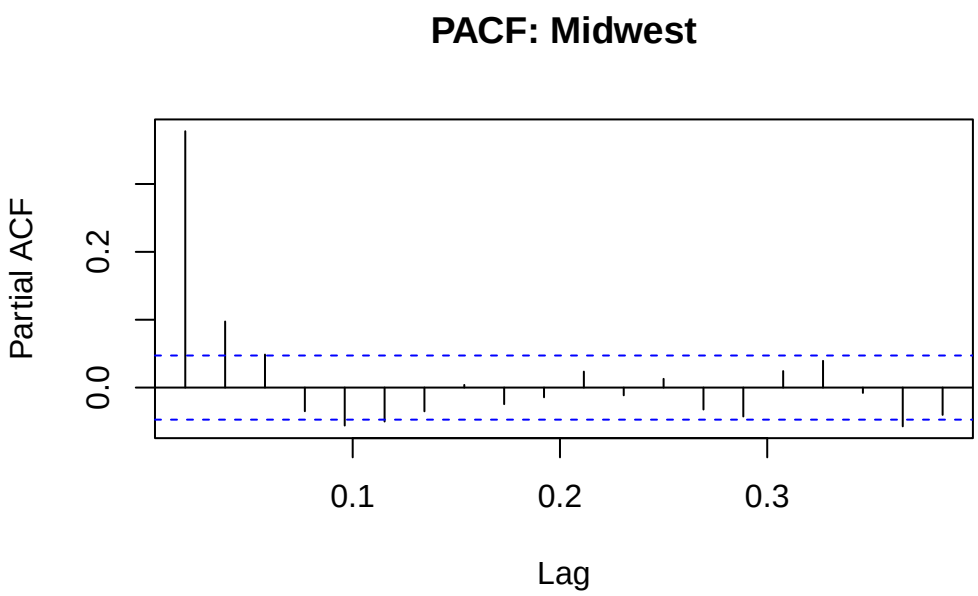


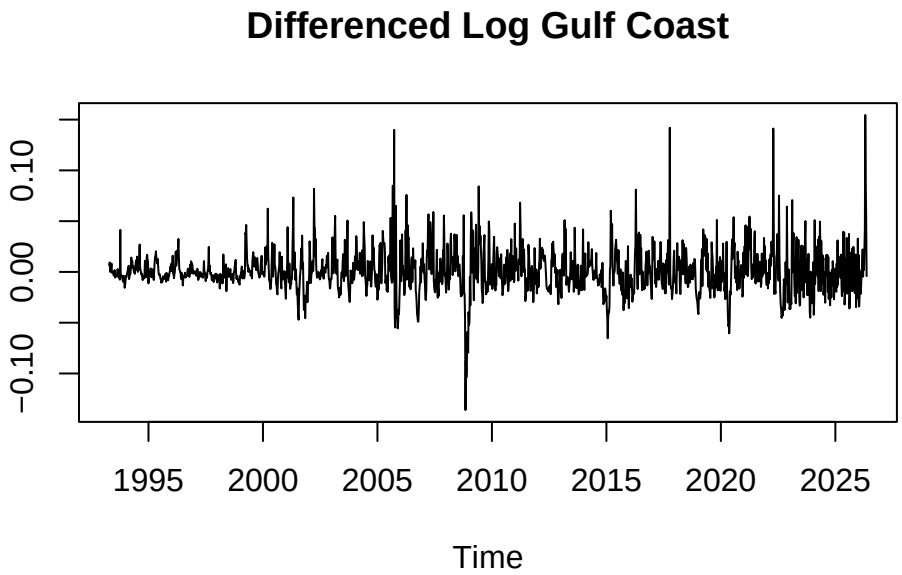

ACF: East Coast
ACF
0.0
0.4
0.8
0.0
0.1
0.2
0.3
0.4
Lag

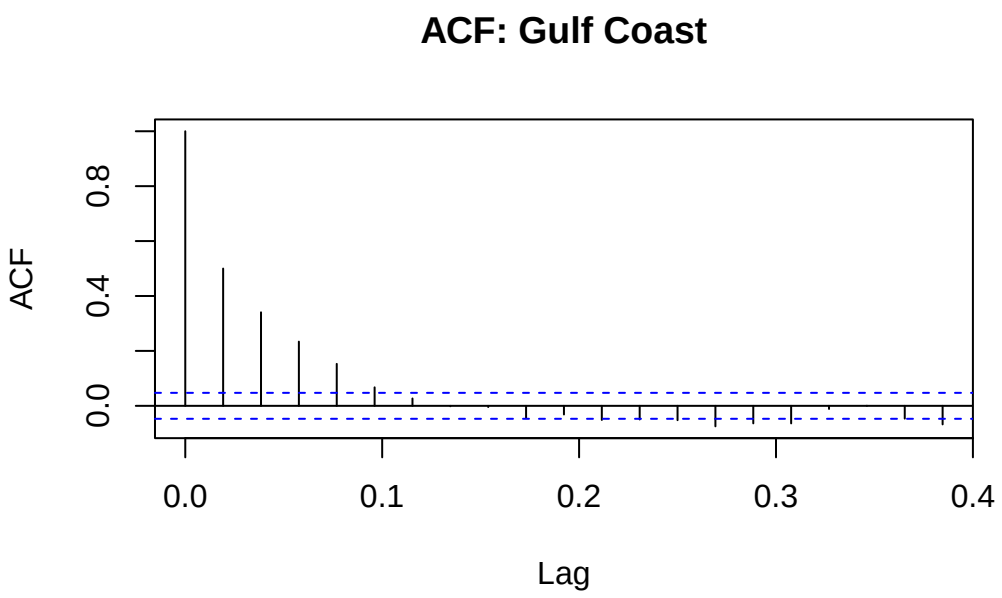

PACF: East Coast
Partial ACF
0.0
0.2
0.4
0.1
0.2
0.3
Lag

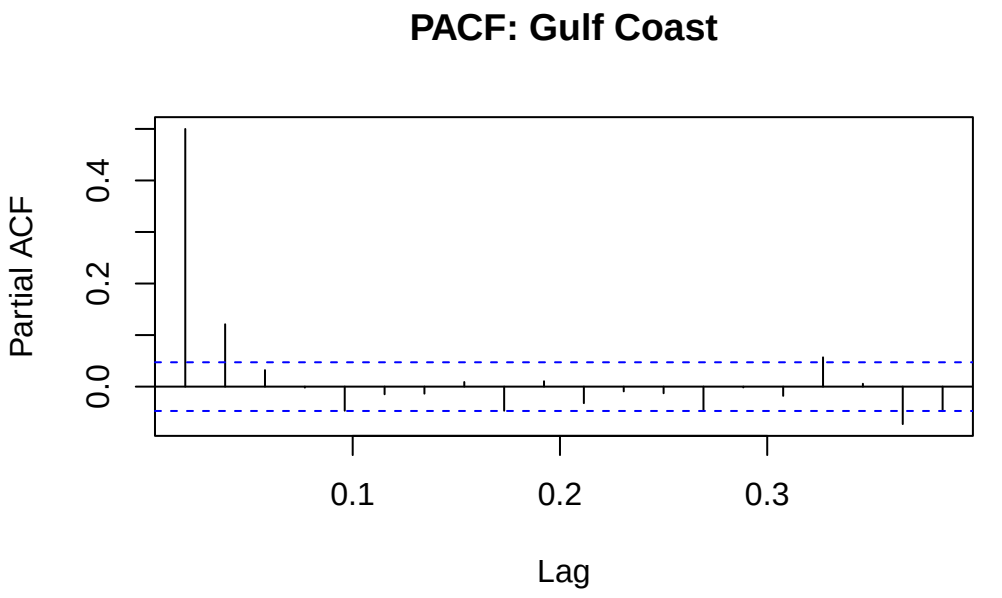

Differenced Log Midwest
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time

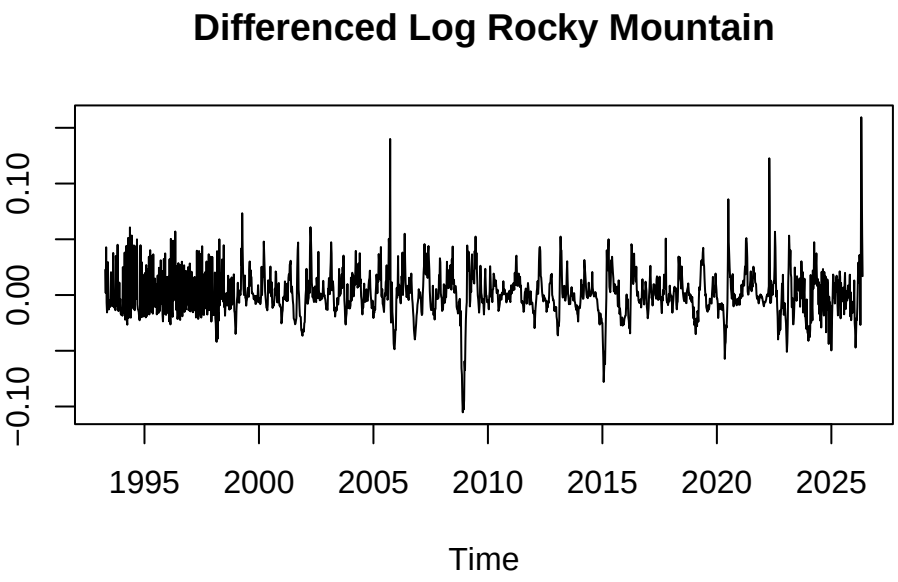

ACF: Midwest
ACF
0.8
0.4
0.0
0.0
0.1
0.2
0.3
0.4
Lag

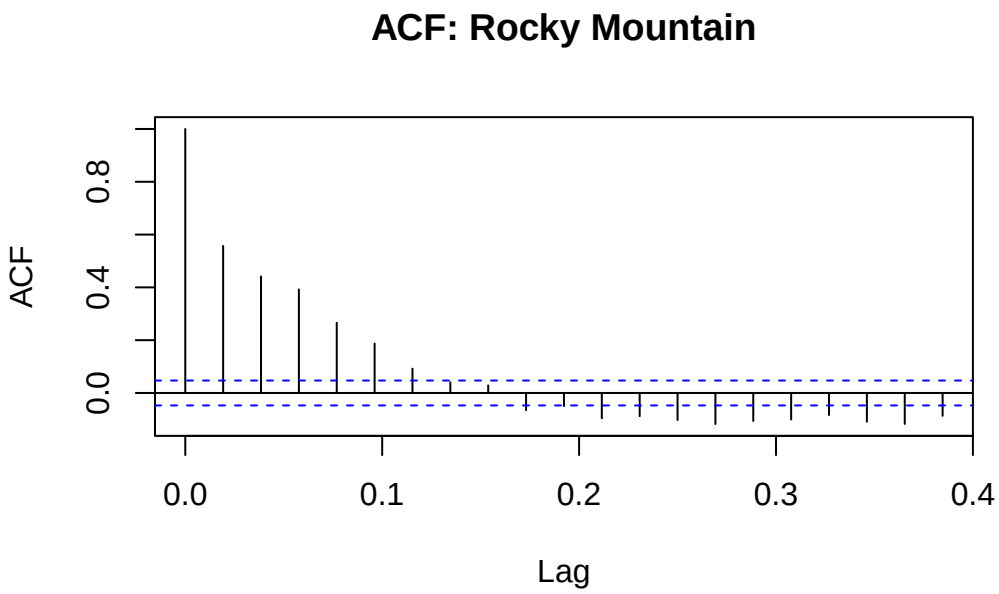

PACF: Midwest
Partial ACF
0.0
0.2
0.1
0.2
0.3
Lag

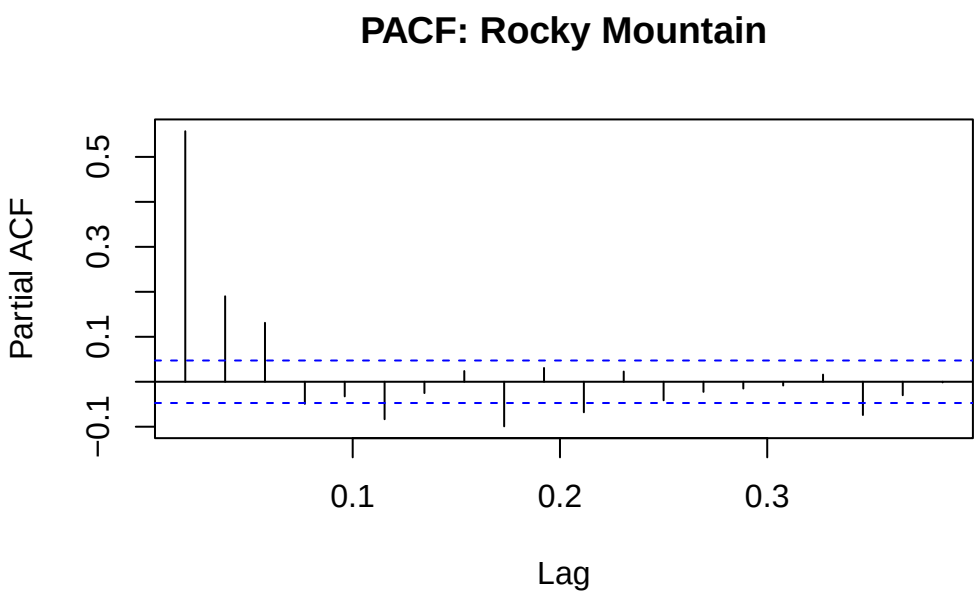

Differenced Log Gulf Coast
−0.10
0.00
0.10
1995
2000
2005
2010
2015
2020
2025
Time

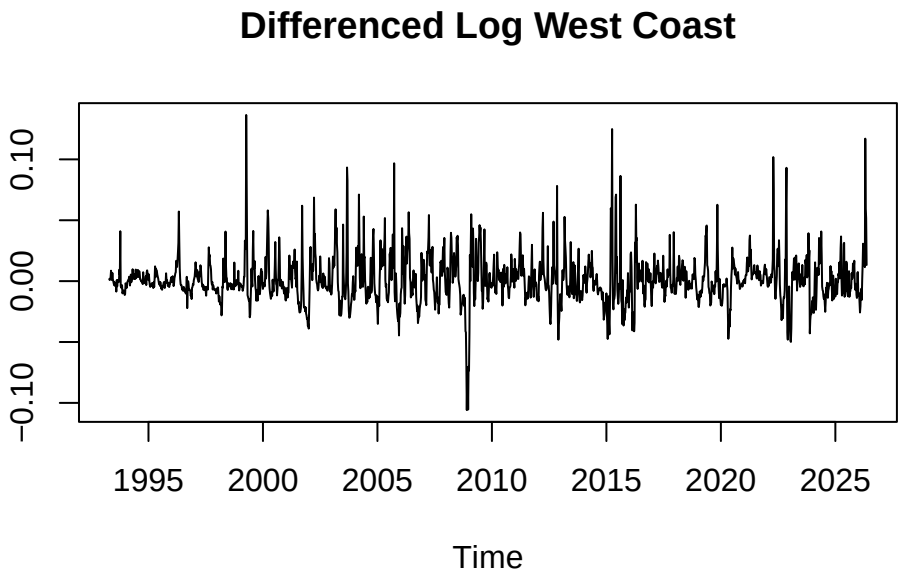

ACF: Gulf Coast
ACF
0.0
0.4
0.8
0.0
0.1
0.2
0.3
0.4
Lag

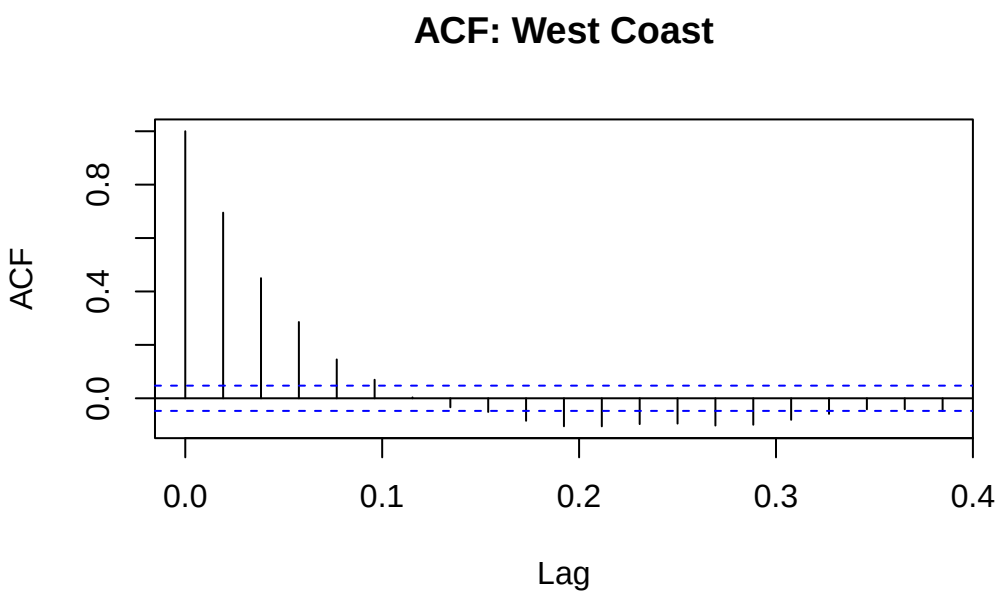

PACF: Gulf Coast
Partial ACF
0.0
0.2
0.4
0.1
0.2
0.3
Lag

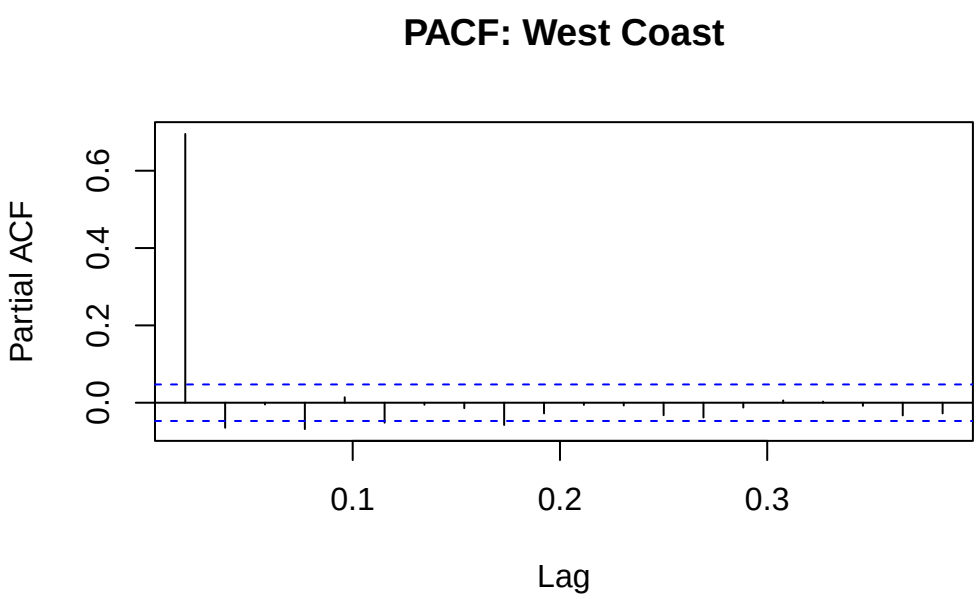

Differenced Log Rocky Mountain
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time


ACF: Rocky Mountain
ACF
0.8
0.4
0.0
0.0
0.1
0.2
0.3
0.4
Lag


PACF: Rocky Mountain
Partial ACF
0.5
0.3
0.1
−0.1
0.1
0.2
0.3
Lag


Differenced Log West Coast
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time





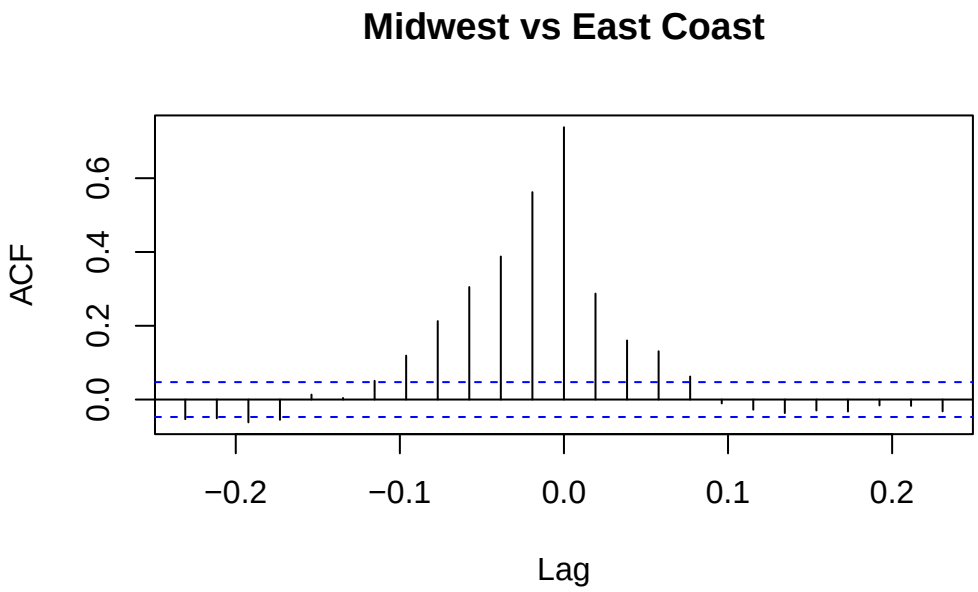

```
par(mfrow = c(1, 1))
```

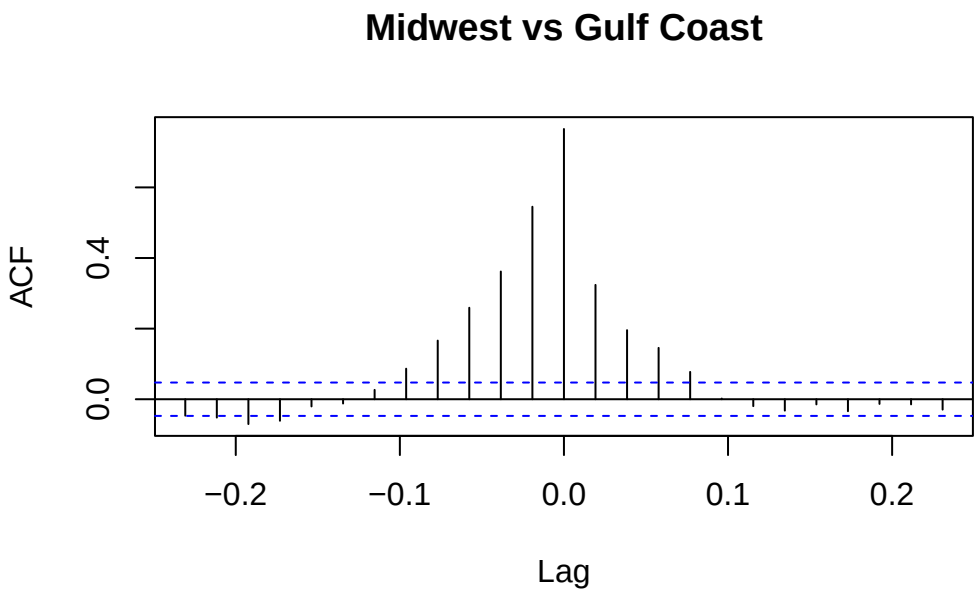
The differenced log models all seem stationary. From the plots of the differenced log models, they seem to be mean zero with no heteroskedasticity and no linear trend. The ACFs tend to

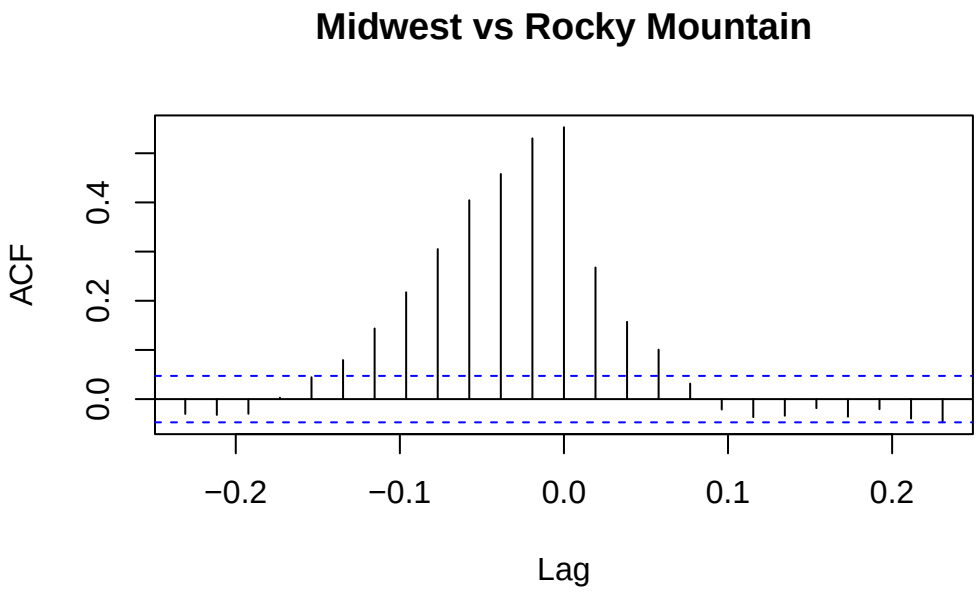
decay quickly to zero, and the PACFs only have a spike close lag 0. There is no evidence of nonstationarity.

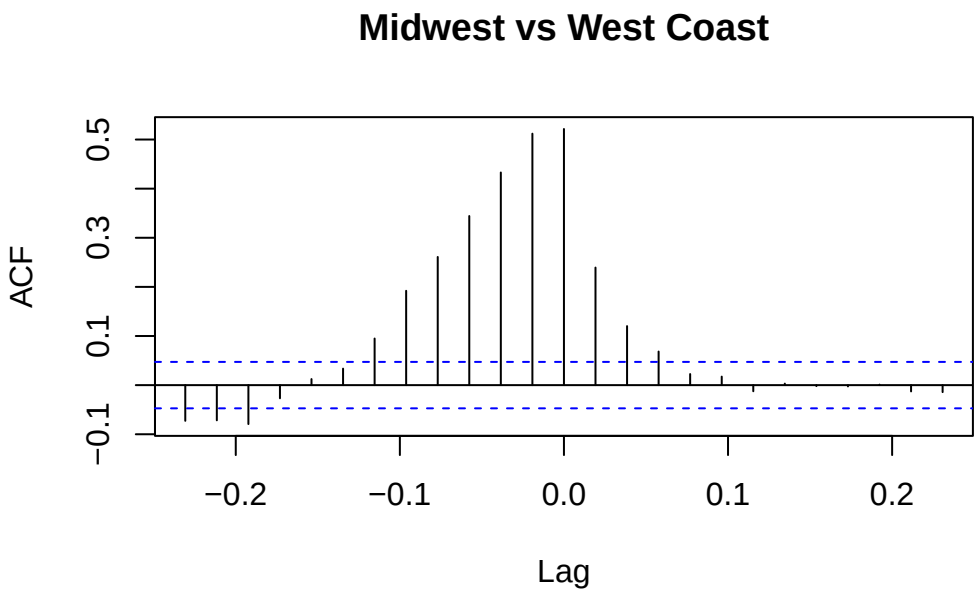
(ii) CCF to understand lag dependence

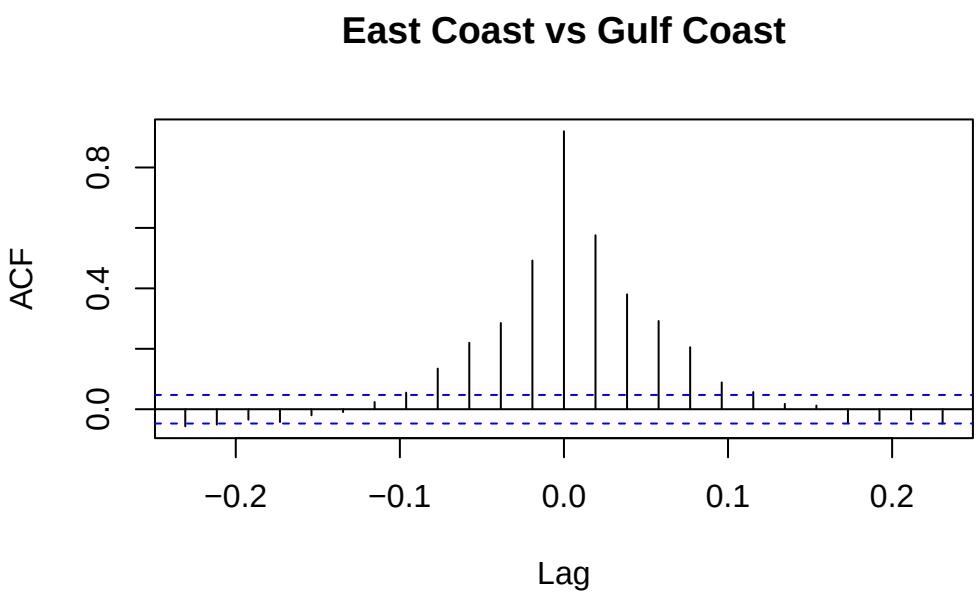
```
# Pairwise CCFs for all 10 regional pairs
region_pairs <- combn(colnames(dRegions), 2, simplify = FALSE)

# Put pairs involving Midwest first
region_pairs <- c(
  Filter(function(x) "Midwest" %in% x, region_pairs),
  Filter(function(x) !("Midwest" %in% x), region_pairs)
)

# Reorder within each pair so Midwest is first
region_pairs <- lapply(region_pairs, function(pair) {
  if ("Midwest" %in% pair) {
    c("Midwest", setdiff(pair, "Midwest"))
  } else {
    pair
  }
})

for (pair in region_pairs) {
  ccf(dRegions[, pair[1]], dRegions[, pair[2]],
      lag.max = 12,
      main = paste(gsub("_", " ", pair[1]), "vs", gsub("_", " ", pair[2])))
}
```

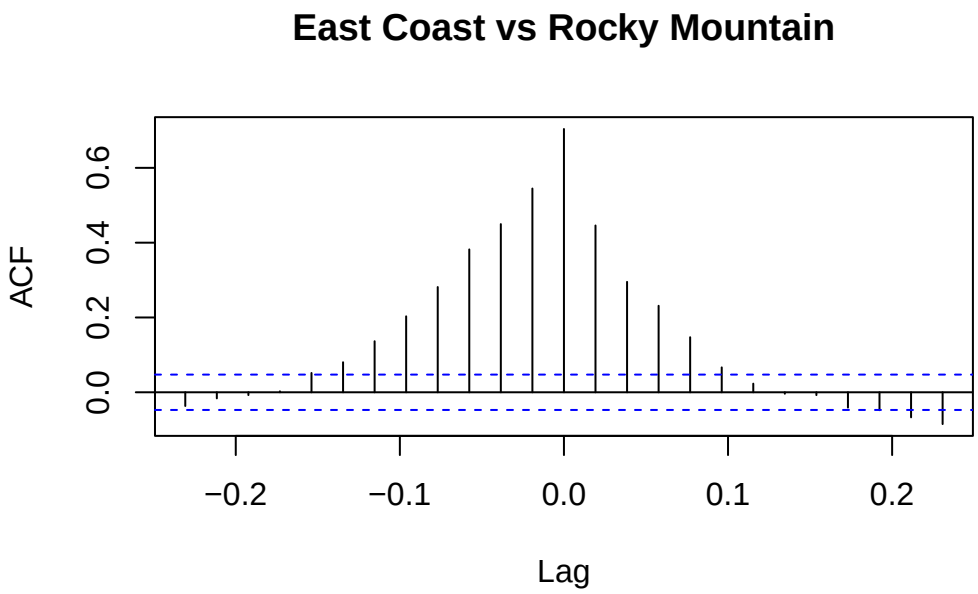

Midwest vs East Coast
ACF
0.0
0.2
0.4
0.6
−0.2
−0.1
0.0
0.1
0.2
Lag

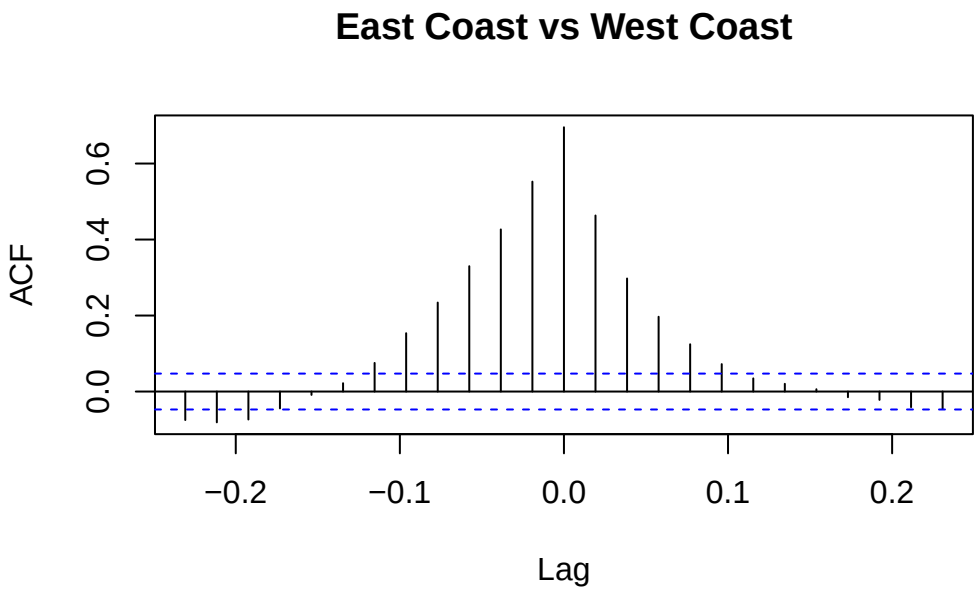

Midwest vs Gulf Coast
ACF
0.0
0.4
−0.2
−0.1
0.0
0.1
0.2
Lag

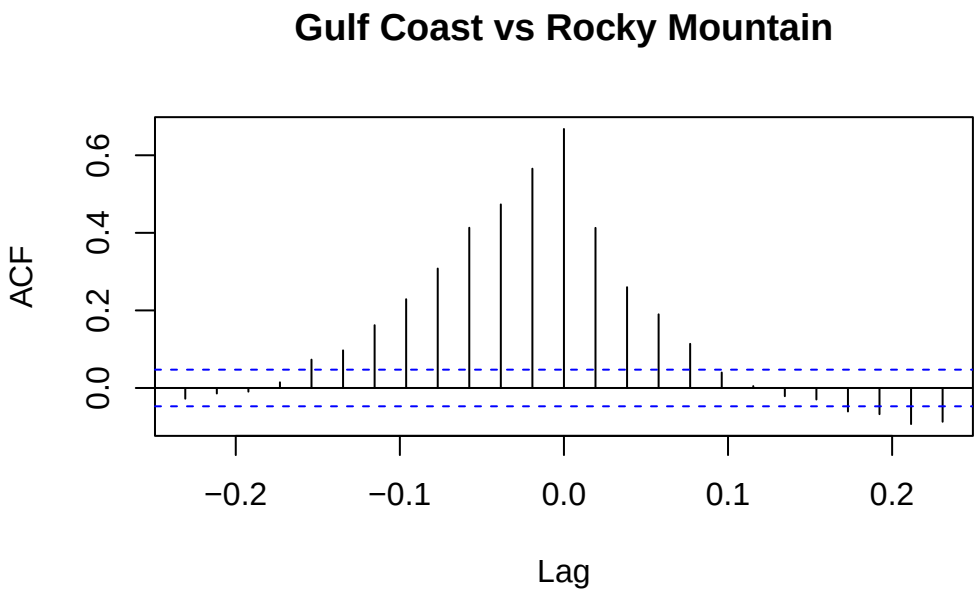

Midwest vs Rocky Mountain
ACF
0.0
0.2
0.4
−0.2
−0.1
0.0
0.1
0.2
Lag

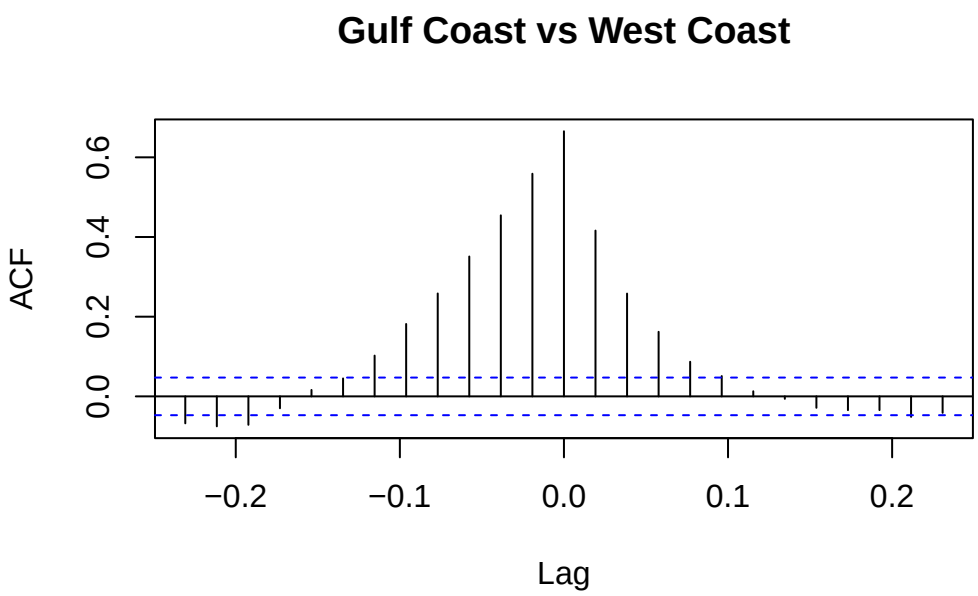

Midwest vs West Coast
ACF
−0.1
0.1
0.3
0.5
−0.2
−0.1
0.0
0.1
0.2
Lag

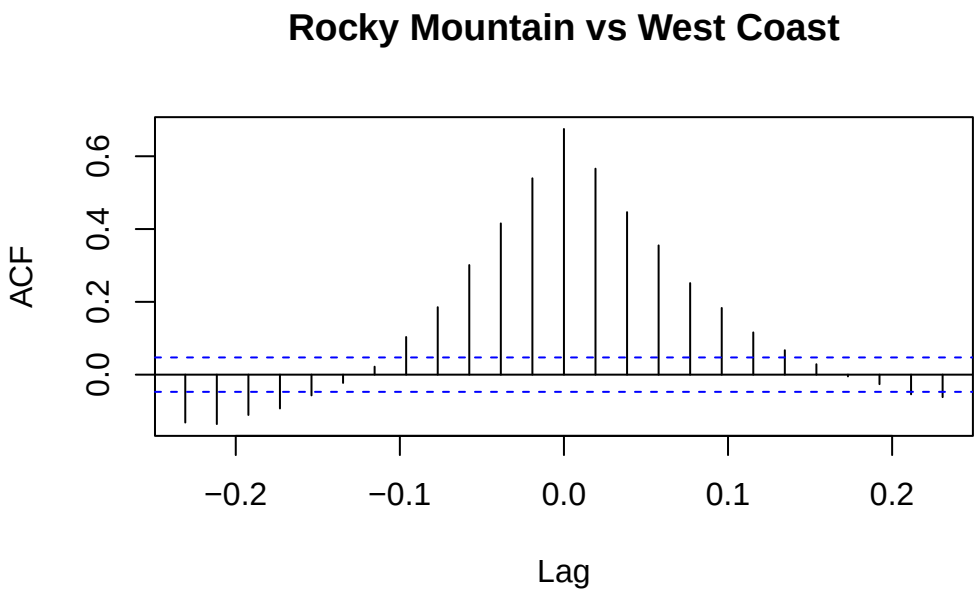

East Coast vs Gulf Coast
ACF
0.0
0.4
0.8
−0.2
−0.1
0.0
0.1
0.2
Lag


East Coast vs Rocky Mountain
ACF
0.0
0.2
0.4
0.6
−0.2
−0.1
0.0
0.1
0.2
Lag


East Coast vs West Coast
ACF
0.0
0.2
0.4
0.6
−0.2
−0.1
0.0
0.1
0.2
Lag


Gulf Coast vs Rocky Mountain
ACF
0.0
0.2
0.4
0.6
−0.2
−0.1
0.0
0.1
0.2
Lag






```
par(mfrow = c(1, 1))
```

Most of the regions are highly correlated with shared shocks (i.e. largest spike at zero). It's

notable that the Midwest does tend to slightly lead the other regions because for the Midwest vs. [region] plots, while the largest spike is at zero, there is a stronger skew to the negative.

(iii) VAR(1) Model

```r
x_regions <- dRegions

# Choose lag length
VARselect(x_regions, lag.max = 8, type = "const")
```

```
$selection
AIC(n)  HQ(n)  SC(n) FPE(n)
     8      3      1      8

$criteria
                   1             2             3             4             5
AIC(n) -4.393438e+01 -4.399915e+01 -4.403971e+01 -4.405313e+01 -4.406199e+01
HQ(n)  -4.389908e+01 -4.393445e+01 -4.394559e+01 -4.392960e+01 -4.390904e+01
SC(n)  -4.383900e+01 -4.382430e+01 -4.378538e+01 -4.371932e+01 -4.364869e+01
FPE(n)  8.308871e-20  7.787730e-20  7.478189e-20  7.378527e-20  7.313512e-20
                   6             7             8
AIC(n) -4.406969e+01 -4.407371e+01 -4.409023e+01
HQ(n)  -4.388733e+01 -4.386194e+01 -4.384905e+01
SC(n)  -4.357691e+01 -4.350145e+01 -4.343849e+01
FPE(n)  7.257477e-20  7.228448e-20  7.110110e-20
```

Tried 1, 3 and 8 but 1 gave best diagnostics.

```r
fit_var_regions <- VAR(x_regions, p = 1, type = "const")
summary(fit_var_regions)
```

```
VAR Estimation Results:
=========================
Endogenous variables: East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast
Deterministic variables: const
Sample size: 1720
Log Likelihood: 25615.901
Roots of the characteristic polynomial:
```

```
0.6515 0.256 0.256 0.198 0.1704
Call:
VAR(y = x_regions, p = 1, type = "const")


Estimation results for equation East_Coast:
===========================================
East_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                   Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     0.0056240  0.0522947   0.108    0.914
Midwest.l1        0.1966443  0.0206092   9.542  < 2e-16 ***
Gulf_Coast.l1     0.2120142  0.0445581   4.758 2.12e-06 ***
Rocky_Mountain.l1 0.0345372  0.0260307   1.327    0.185
West_Coast.l1     0.1164431  0.0274288   4.245 2.30e-05 ***
const             0.0003352  0.0003536   0.948    0.343
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01465 on 1714 degrees of freedom
Multiple R-Squared: 0.3784, Adjusted R-squared: 0.3766
F-statistic: 208.7 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Midwest:
========================================
Midwest = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + co

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.3036740  0.0885660  -3.429 0.000621 ***
Midwest.l1         0.3207444  0.0349036   9.189  < 2e-16 ***
Gulf_Coast.l1      0.2389347  0.0754633   3.166 0.001571 **
Rocky_Mountain.l1  0.1179783  0.0440855   2.676 0.007519 **
West_Coast.l1      0.0415396  0.0464533   0.894 0.371328
const              0.0004301  0.0005989   0.718 0.472775
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.02481 on 1714 degrees of freedom
Multiple R-Squared: 0.1541, Adjusted R-squared: 0.1517
F-statistic: 62.47 on 5 and 1714 DF,  p-value: < 2.2e-16
```

```
Estimation results for equation Gulf_Coast:
===========================================
Gulf_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                   Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     0.0323409  0.0630936   0.513 0.608308
Midwest.l1        0.3041834  0.0248650  12.233  < 2e-16 ***
Gulf_Coast.l1     0.0663471  0.0537594   1.234 0.217317
Rocky_Mountain.l1 0.0660422  0.0314061   2.103 0.035625 *
West_Coast.l1     0.1241504  0.0330929   3.752 0.000182 ***
const             0.0002604  0.0004267   0.610 0.541722
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01767 on 1714 degrees of freedom
Multiple R-Squared: 0.327,  Adjusted R-squared: 0.325
F-statistic: 166.6 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Rocky_Mountain:
===============================================
Rocky_Mountain = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.1804089  0.0555926  -3.245   0.0012 **
Midwest.l1         0.1668014  0.0219089   7.613 4.38e-14 ***
Gulf_Coast.l1      0.2111636  0.0473681   4.458 8.81e-06 ***
Rocky_Mountain.l1  0.2225497  0.0276723   8.042 1.63e-15 ***
West_Coast.l1      0.2856227  0.0291586   9.795  < 2e-16 ***
const              0.0001872  0.0003759   0.498   0.6186
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01557 on 1714 degrees of freedom
Multiple R-Squared: 0.428,  Adjusted R-squared: 0.4263
F-statistic: 256.5 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation West_Coast:
```

```
=========================================
West_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.1479101  0.0478854  -3.089  0.00204 **
Midwest.l1         0.1370713  0.0188715   7.263  5.7e-13 ***
Gulf_Coast.l1      0.1048158  0.0408012   2.569  0.01028 *
Rocky_Mountain.l1  0.0684224  0.0238359   2.871  0.00415 **
West_Coast.l1      0.5673976  0.0251161  22.591  < 2e-16 ***
const              0.0002801  0.0003238   0.865  0.38719
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01341 on 1714 degrees of freedom
Multiple R-Squared: 0.5184, Adjusted R-squared: 0.517
F-statistic:   369 on 5 and 1714 DF,  p-value: < 2.2e-16



Covariance matrix of residuals:
               East_Coast   Midwest Gulf_Coast Rocky_Mountain West_Coast
East_Coast      0.0002146 0.0002555  0.0002291      1.226e-04  1.092e-04
Midwest         0.0002555 0.0006154  0.0003188      1.786e-04  1.548e-04
Gulf_Coast      0.0002291 0.0003188  0.0003123      1.397e-04  1.276e-04
Rocky_Mountain  0.0001226 0.0001786  0.0001397      2.425e-04  8.887e-05
West_Coast      0.0001092 0.0001548  0.0001276      8.887e-05  1.799e-04

Correlation matrix of residuals:
               East_Coast Midwest Gulf_Coast Rocky_Mountain West_Coast
East_Coast         1.0000  0.7031     0.8850         0.5377     0.5557
Midwest            0.7031  1.0000     0.7271         0.4624     0.4653
Gulf_Coast         0.8850  0.7271     1.0000         0.5075     0.5383
Rocky_Mountain     0.5377  0.4624     0.5075         1.0000     0.4255
West_Coast         0.5557  0.4653     0.5383         0.4255     1.0000
```

Since all roots are less than 1, the model is stable.

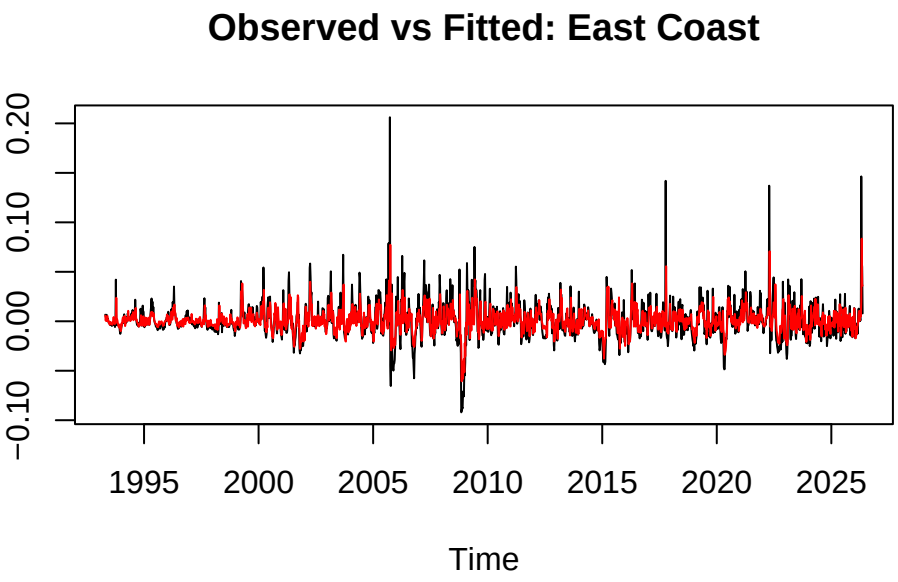
NOTE: finish evaluations of coefficients, notice both significance as well as magnitude of coefficients

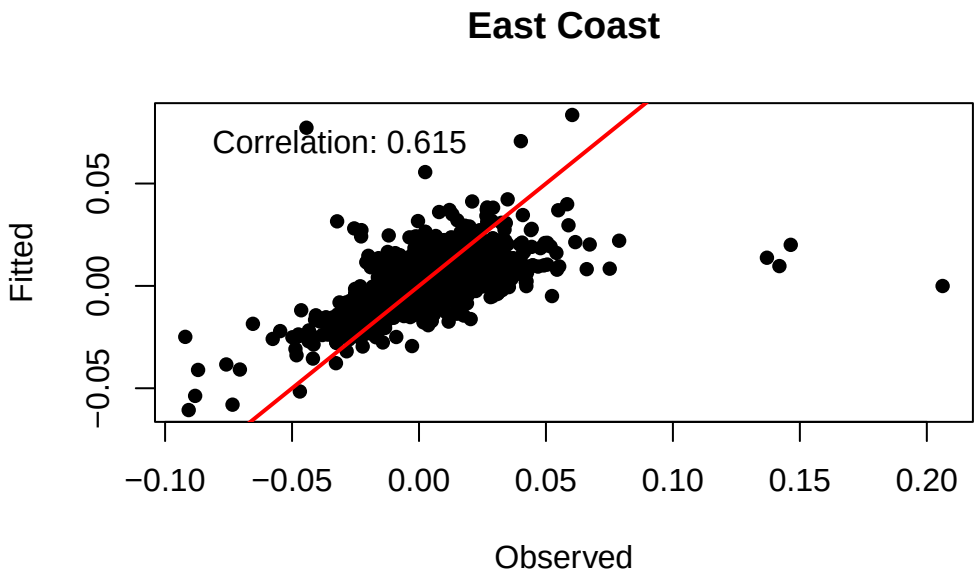
Correlation pattern of residuals: higher correlation means they are affected by the same shocks, whereas lower correlation means they are affected by more local dynamics. **FINISH**

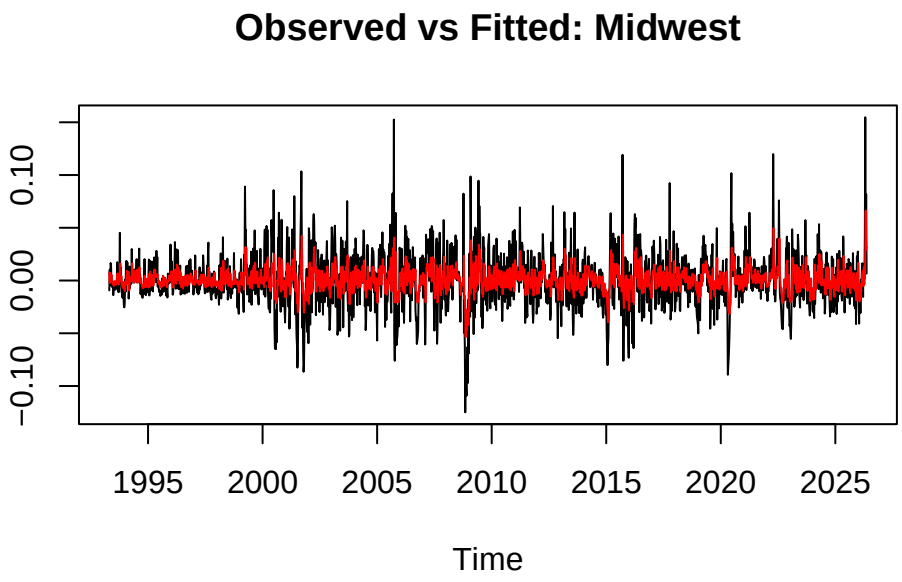
(iv) Diagnostics

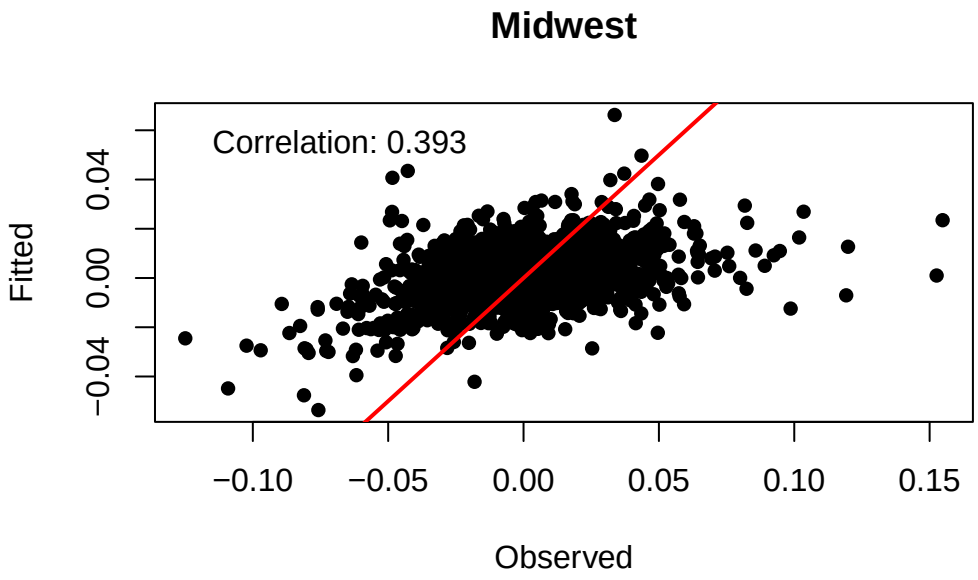
```r
# Fitted values and residuals
fitted_regions <- fitted(fit_var_regions)
res_regions <- resid(fit_var_regions)
```

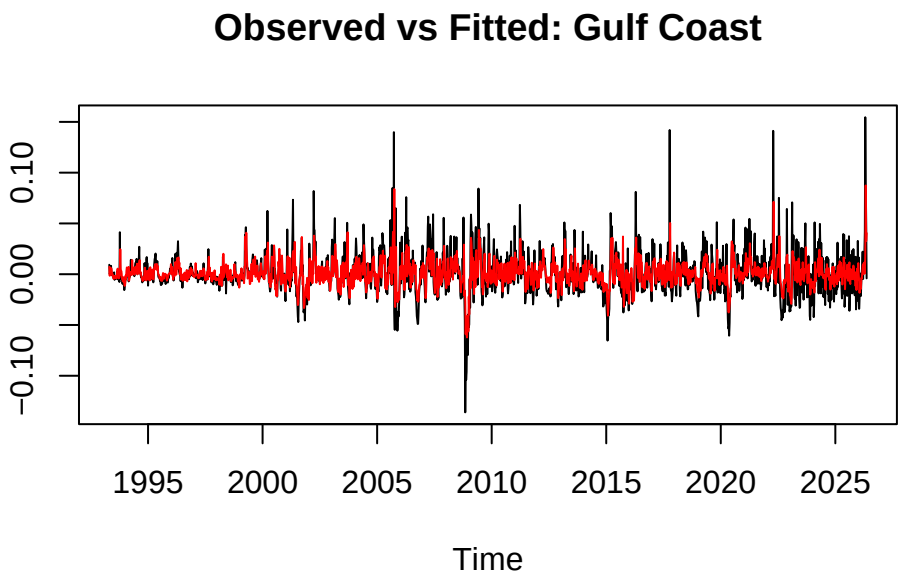
```r
# Observed vs fitted plots for each region

p_used <- fit_var_regions$p

for (nm in colnames(x_regions)) {
  obs <- window(x_regions[, nm], start = time(x_regions)[p_used + 1])
  fit <- ts(fitted_regions[, nm],
            start = start(obs),
            frequency = frequency(obs))

  plot(obs, type = "l", col = "black",
       main = paste("Observed vs Fitted:", gsub("_", " ", nm)),
       ylab = "", xlab = "Time")
  lines(fit, col = "red")

  plot(obs, fit,
       pch = 16,
       xlab = "Observed",
       ylab = "Fitted",
       main = gsub("_", " ", nm))
  abline(0, 1, col = "red", lwd = 2)
  legend("topleft",
       legend = paste("Correlation:",
                      round(cor(obs, fit), 3)),
       bty = "n")

}
```

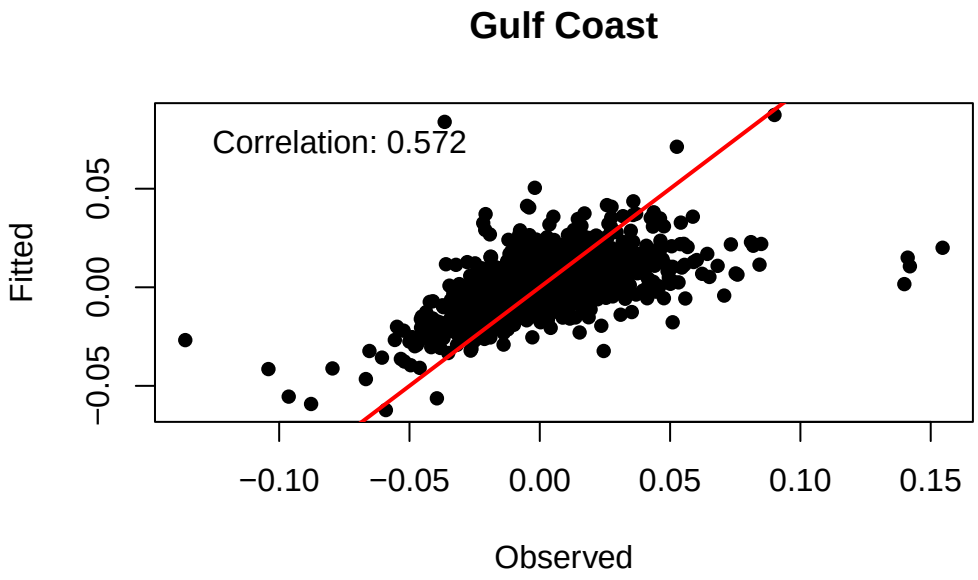

Observed vs Fitted: East Coast
0.20
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time

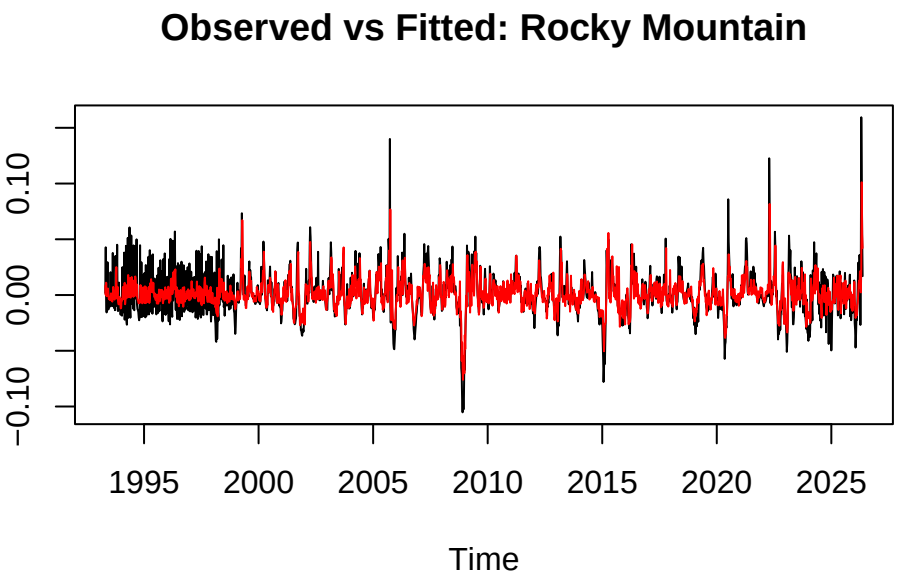

East Coast
Correlation: 0.615
Fitted
0.05
0.00
−0.05
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Observed

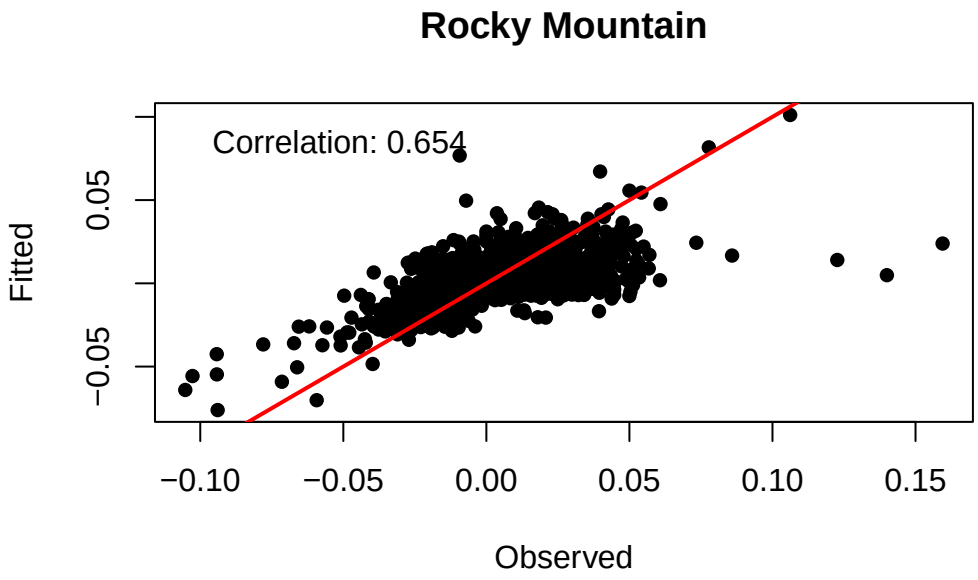

Observed vs Fitted: Midwest
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time

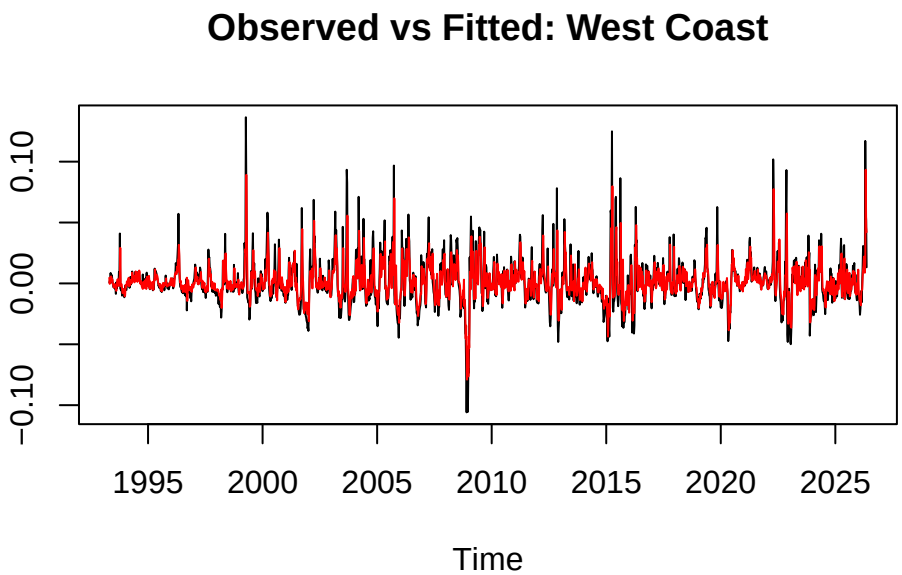

Midwest
Correlation: 0.393
Fitted
0.04
0.00
−0.04
−0.10
−0.05
0.00
0.05
0.10
0.15
Observed

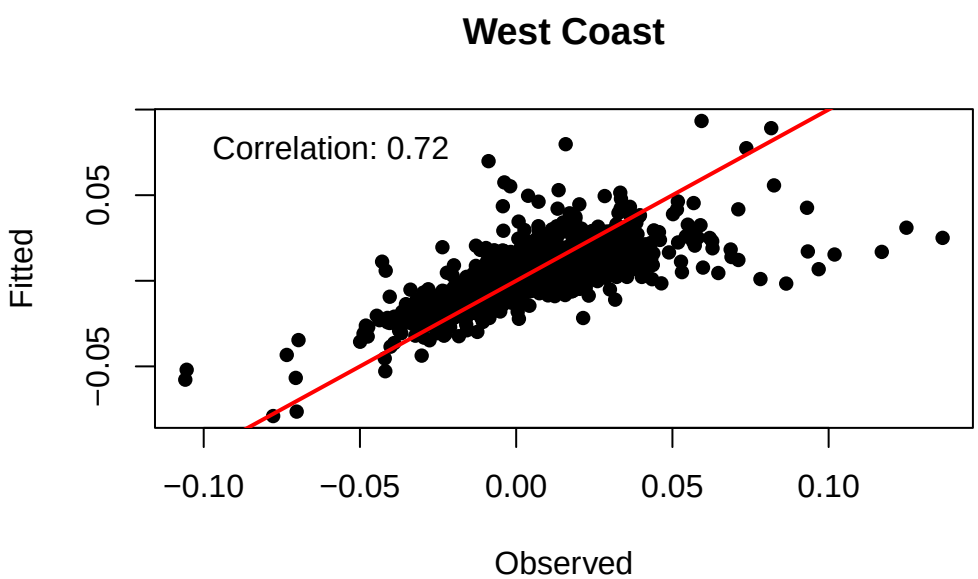

Observed vs Fitted: Gulf Coast
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time


Gulf Coast
Correlation: 0.572
Fitted
0.05
0.00
−0.05
−0.10
−0.05
0.00
0.05
0.10
0.15
Observed


Observed vs Fitted: Rocky Mountain
0.10
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time


Rocky Mountain
Correlation: 0.654
Fitted
0.05
−0.05
−0.10
−0.05
0.00
0.05
0.10
0.15
Observed

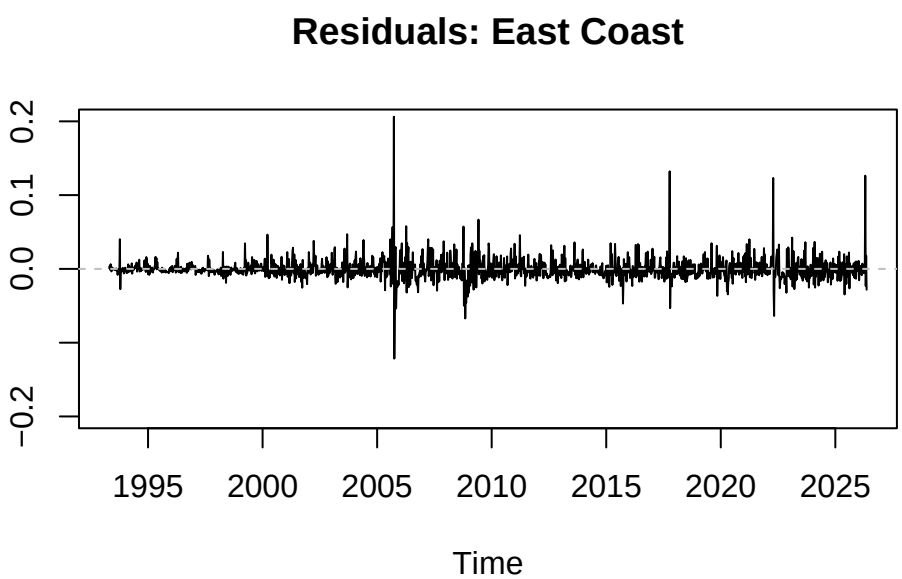


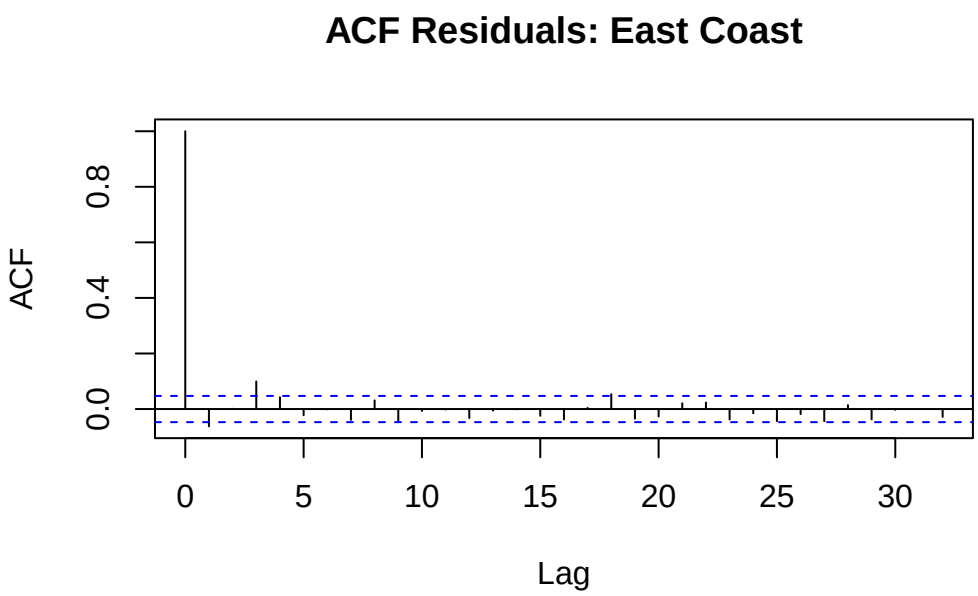


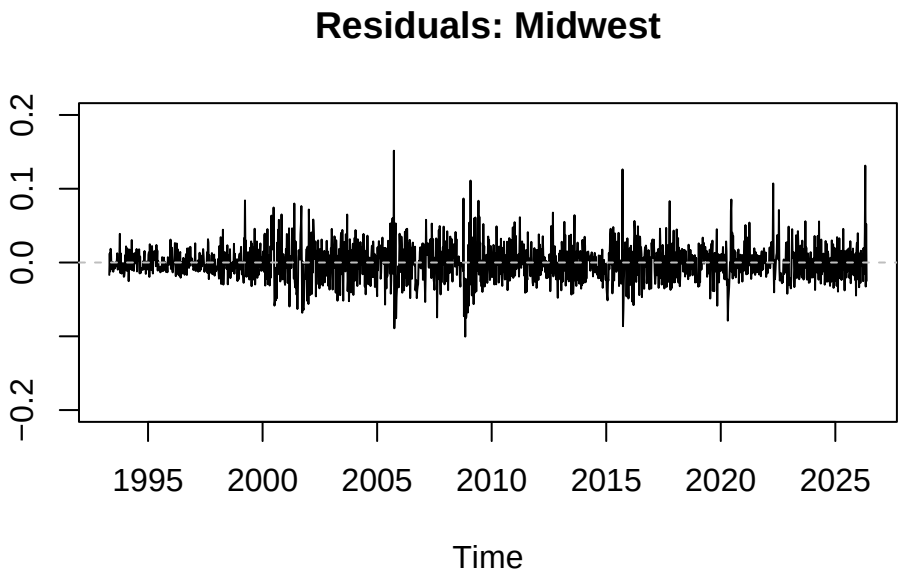

```
par(mfrow = c(1, 1))
```

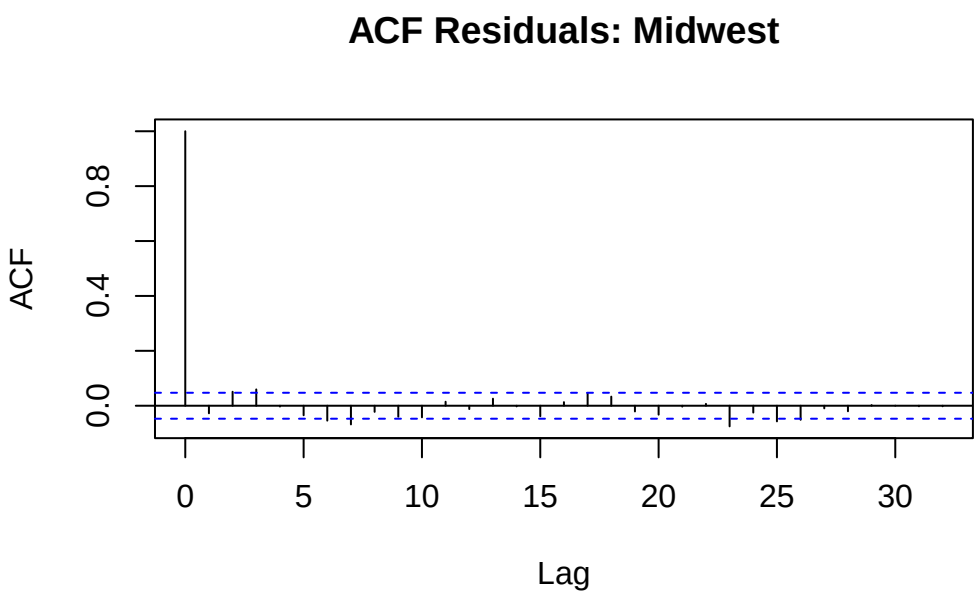
Does good job everywhere except pre 2000s for rocky mountain. Midwest captures mean structure but no shocks.

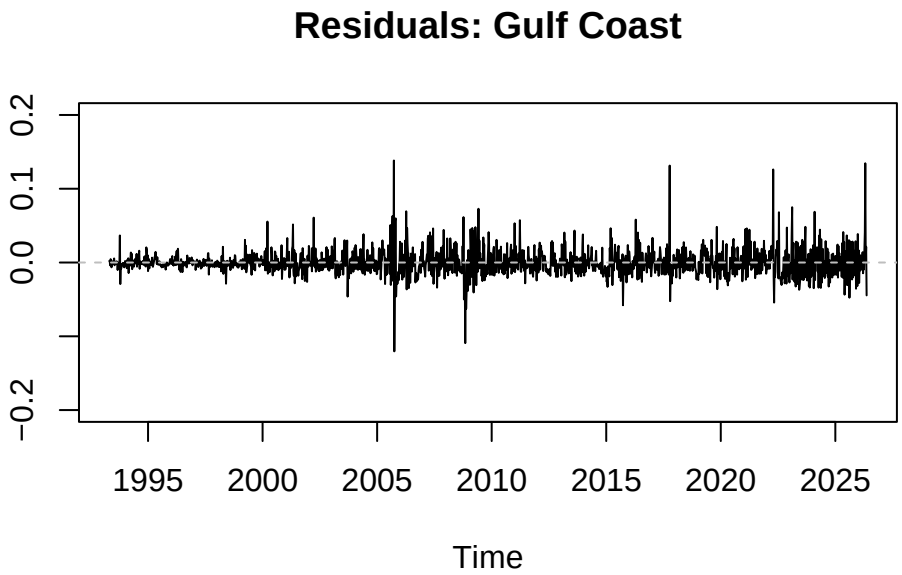
```r
# Residual series and residual ACFs

for (nm in colnames(res_regions)) {
  plot(ts(res_regions[, nm],
          start = time(x_regions)[p_used + 1],
          frequency = frequency(x_regions)),
       type = "l",
       main = paste("Residuals:", gsub("_", " ", nm)),
       ylab = "", xlab = "Time", ylim = c(-0.2,0.2))
  abline(h = 0, lty = 2, col = "gray")

  acf(res_regions[, nm],
      main = paste("ACF Residuals:", gsub("_", " ", nm)))
}
```

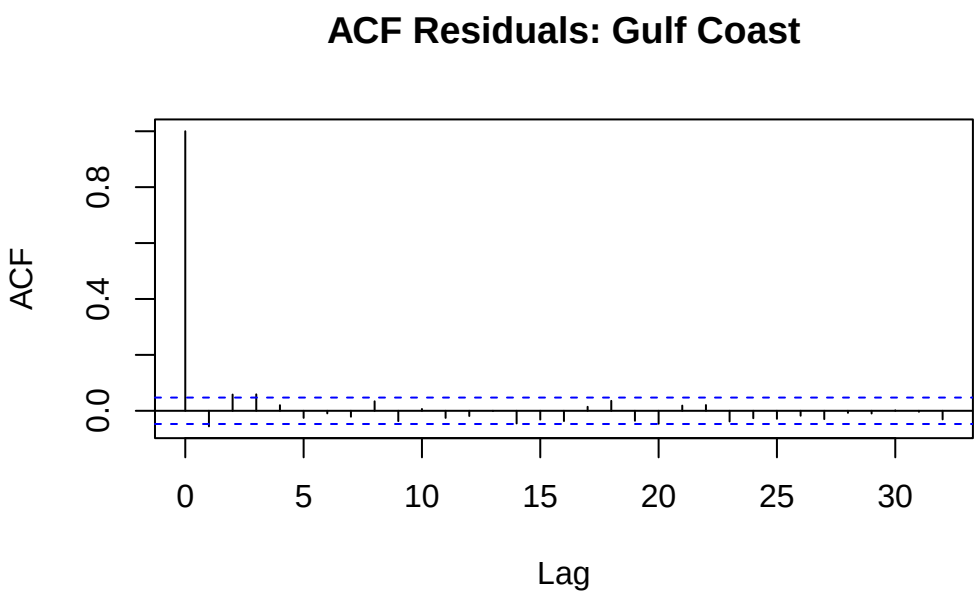


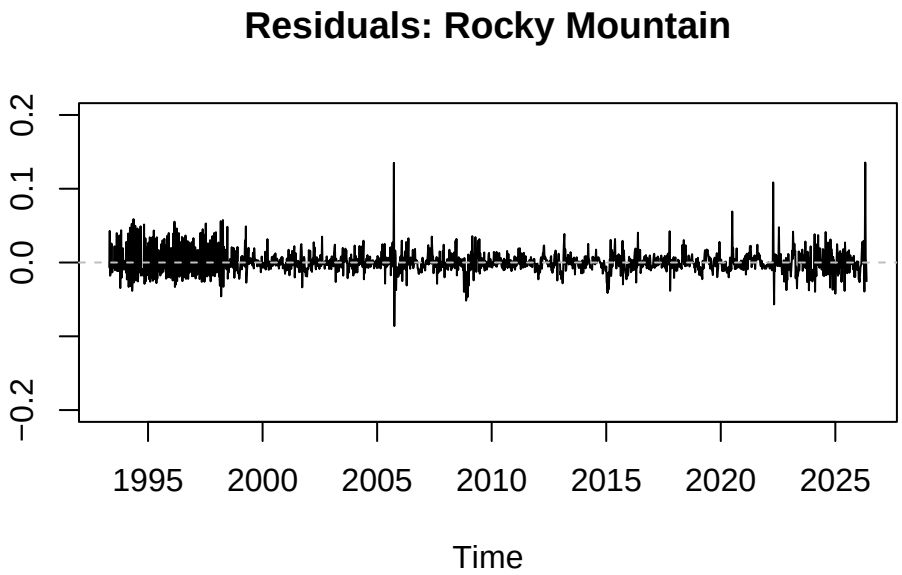

ACF Residuals: East Coast
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag

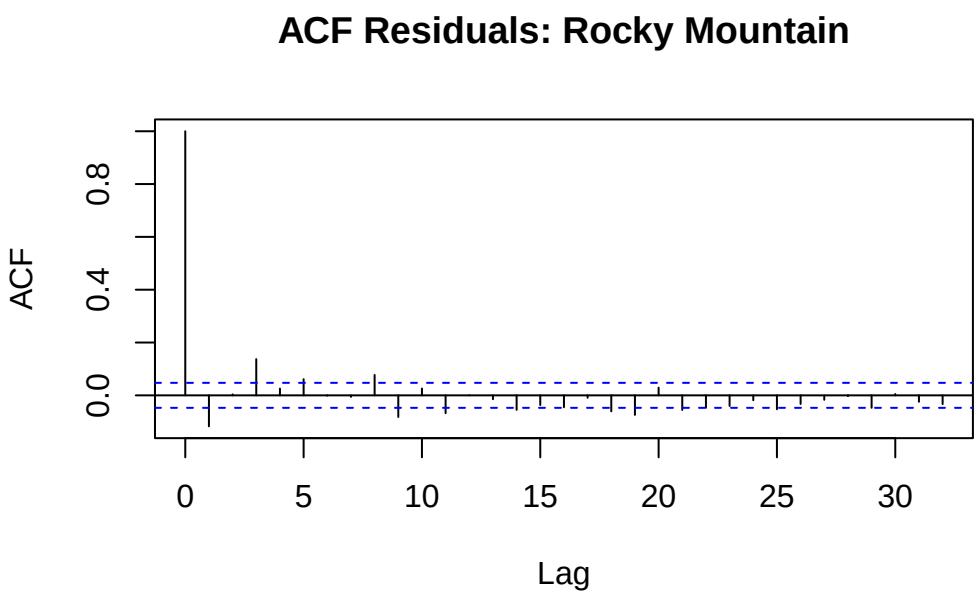

Residuals: Midwest
−0.2
0.0
0.1
0.2
1995
2000
2005
2010
2015
2020
2025
Time

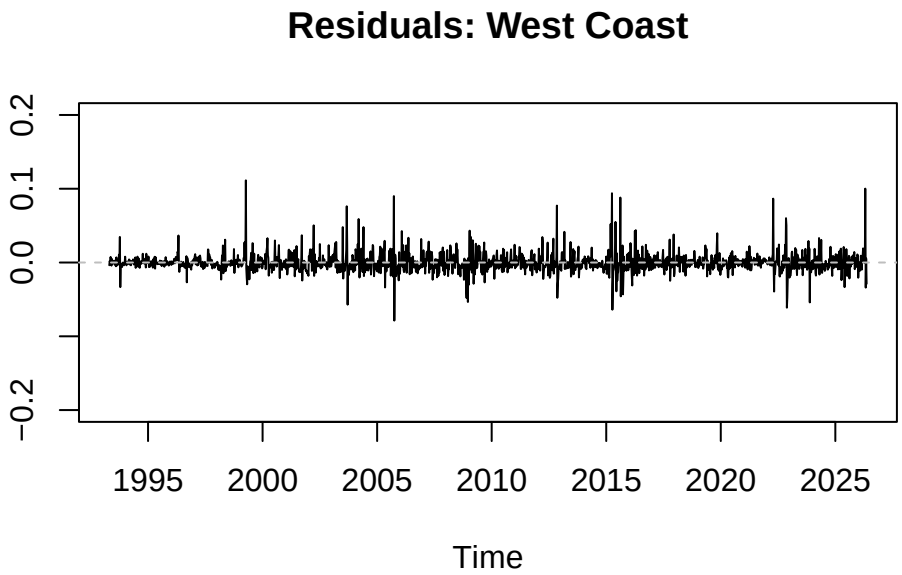

ACF Residuals: Midwest
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag

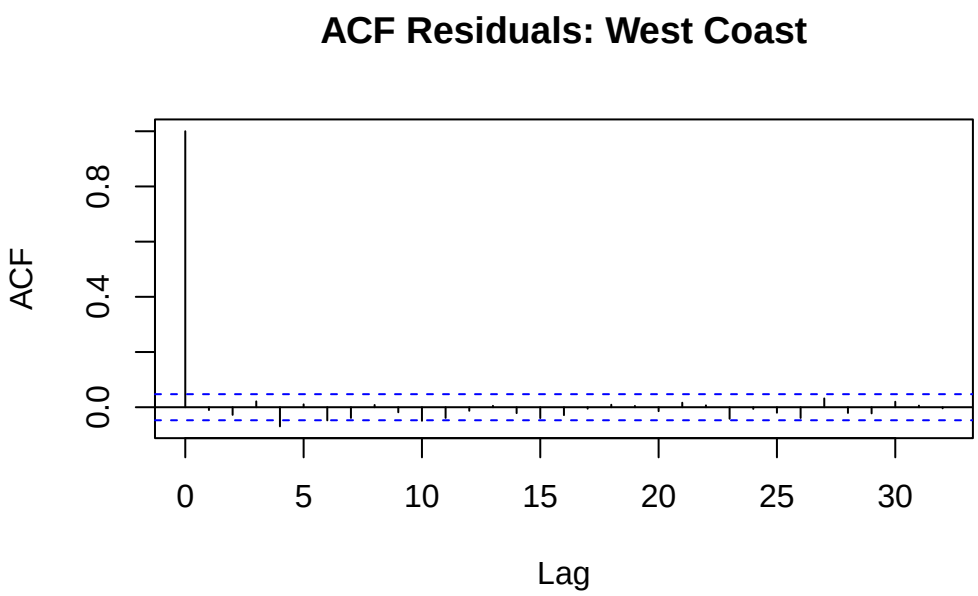

Residuals: Gulf Coast
−0.2
0.0
0.1
0.2
1995
2000
2005
2010
2015
2020
2025
Time


ACF Residuals: Gulf Coast
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag


Residuals: Rocky Mountain
−0.2
0.0
0.1
0.2
1995
2000
2005
2010
2015
2020
2025
Time


ACF Residuals: Rocky Mountain
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag


Residuals: West Coast
−0.2
0.0
0.1
0.2
1995
2000
2005
2010
2015
2020
2025
Time




```
par(mfrow = c(1, 1))
```

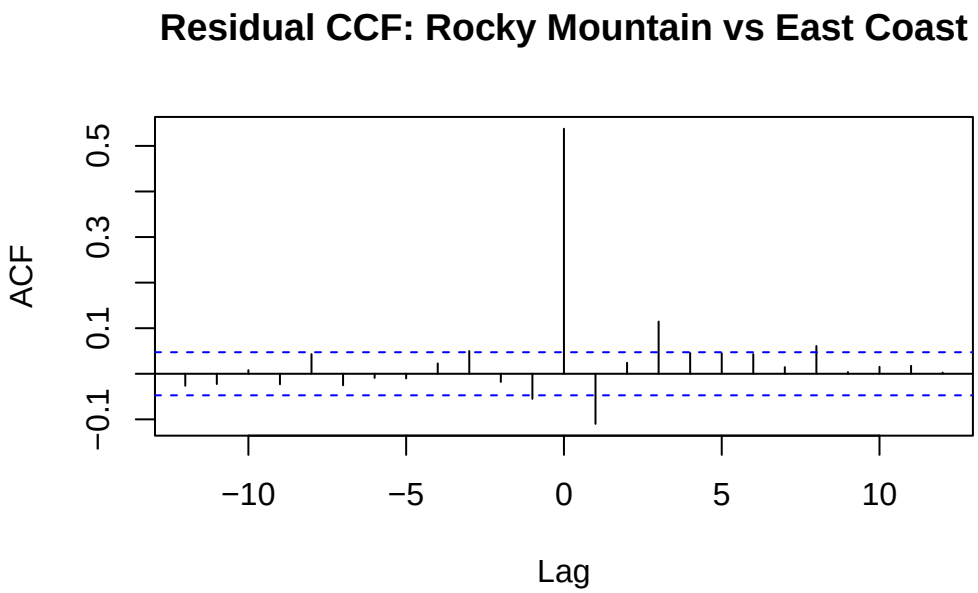
Residuals are largely white noise except Rocky Mountain has a slight decay in ACF.

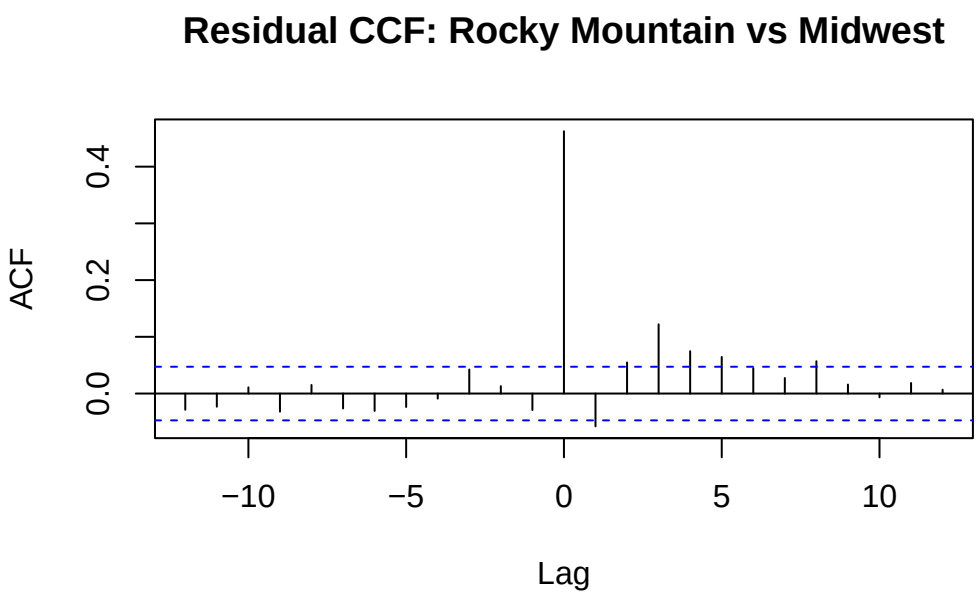
FINISH: which spikes and are they at the same time?

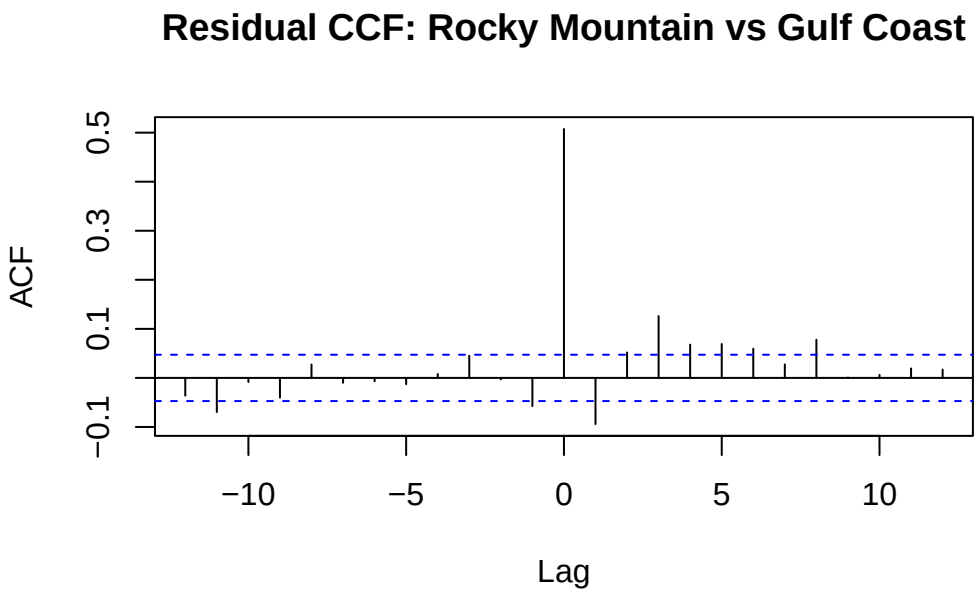
again rocky mountain weird

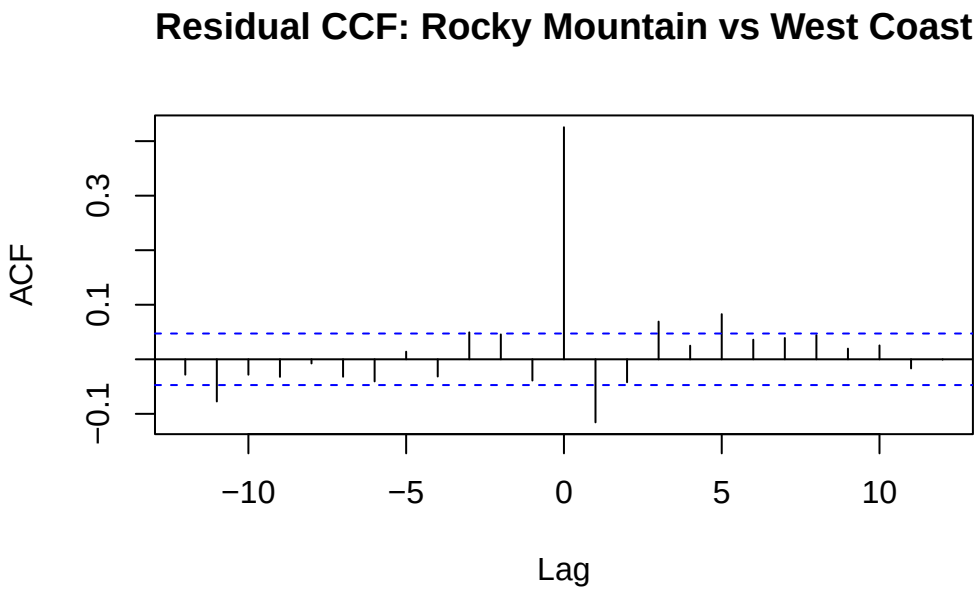
```
# Residual CCFs for all regional pairs
region_pairs <- combn(colnames(res_regions), 2, simplify = FALSE)

# Put pairs involving Rocky Mountain first
region_pairs <- c(
  Filter(function(x) "Rocky_Mountain" %in% x, region_pairs),
  Filter(function(x) !("Rocky_Mountain" %in% x), region_pairs)
)

# Reorder within each pair so Rocky Mountain is first
region_pairs <- lapply(region_pairs, function(pair) {
  if ("Rocky_Mountain" %in% pair) {
    c("Rocky_Mountain", setdiff(pair, "Rocky_Mountain"))
  } else {
    pair
```

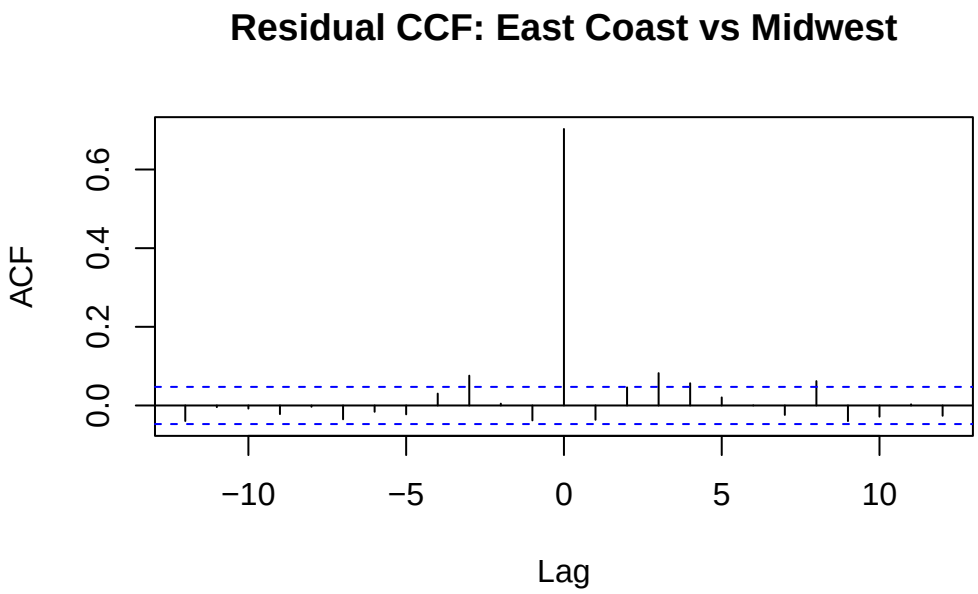
```
  }
})

for (pair in region_pairs) {
  ccf(res_regions[, pair[1]], res_regions[, pair[2]],
      lag.max = 12,
      main = paste("Residual CCF:",
                   gsub("_", " ", pair[1]), "vs", gsub("_", " ", pair[2])))
}
```

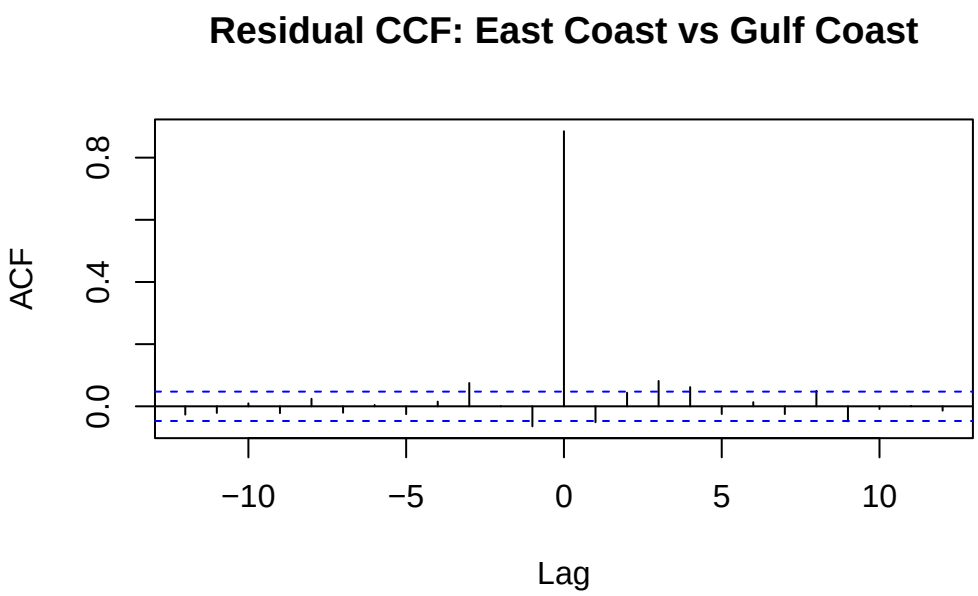


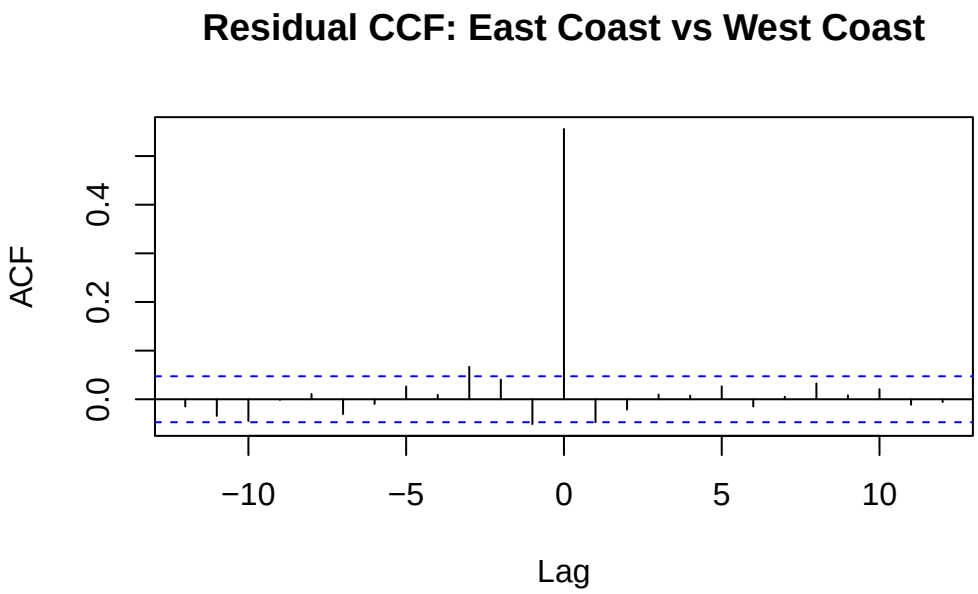

Residual CCF: Rocky Mountain vs Midwest
ACF
0.0
0.2
0.4
−10
−5
0
5
10
Lag

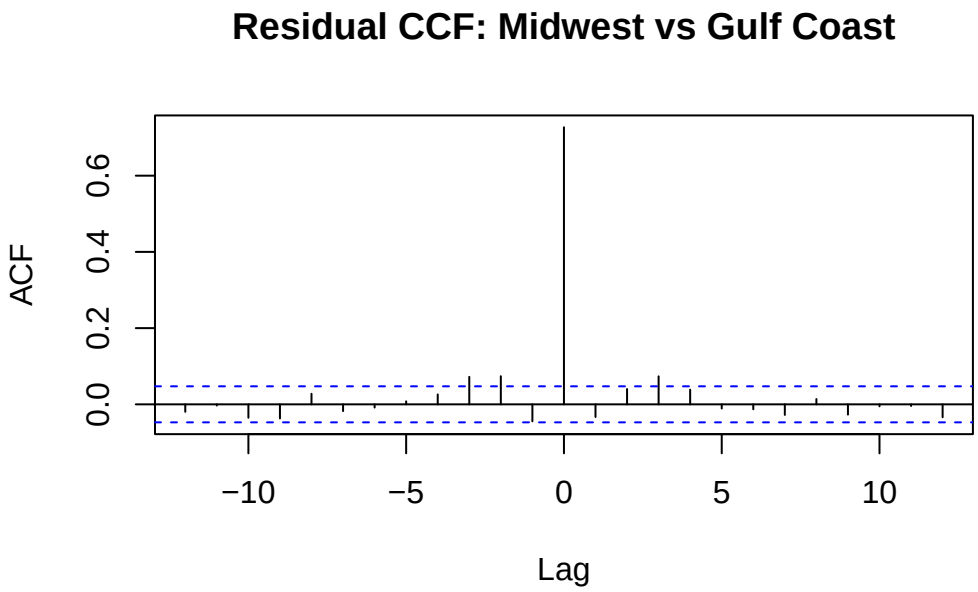

Residual CCF: Rocky Mountain vs Gulf Coast
ACF
−0.1
0.1
0.3
0.5
−10
−5
0
5
10
Lag

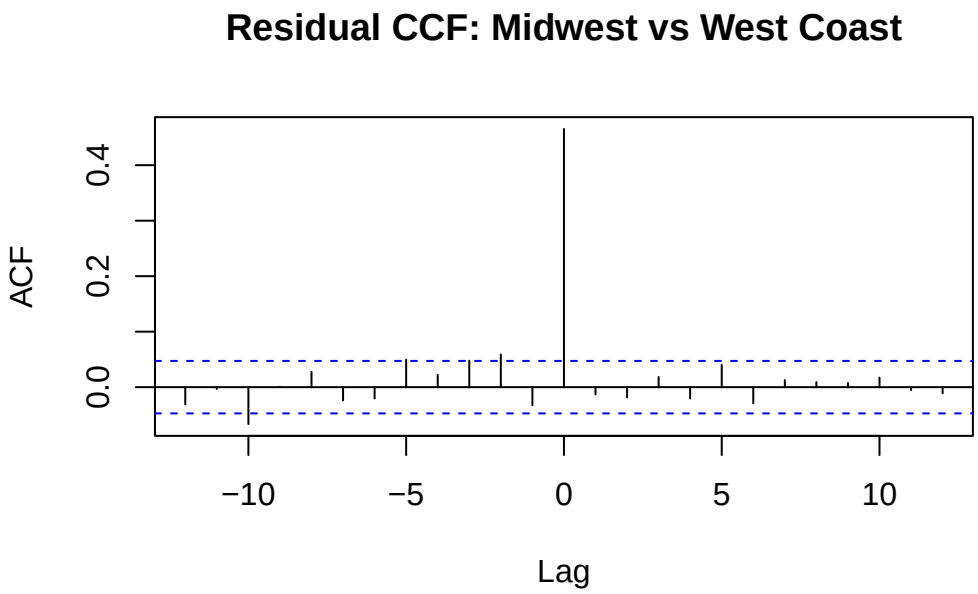

Residual CCF: Rocky Mountain vs West Coast
ACF
0.3
0.1
−0.1
−10
−5
0
5
10
Lag

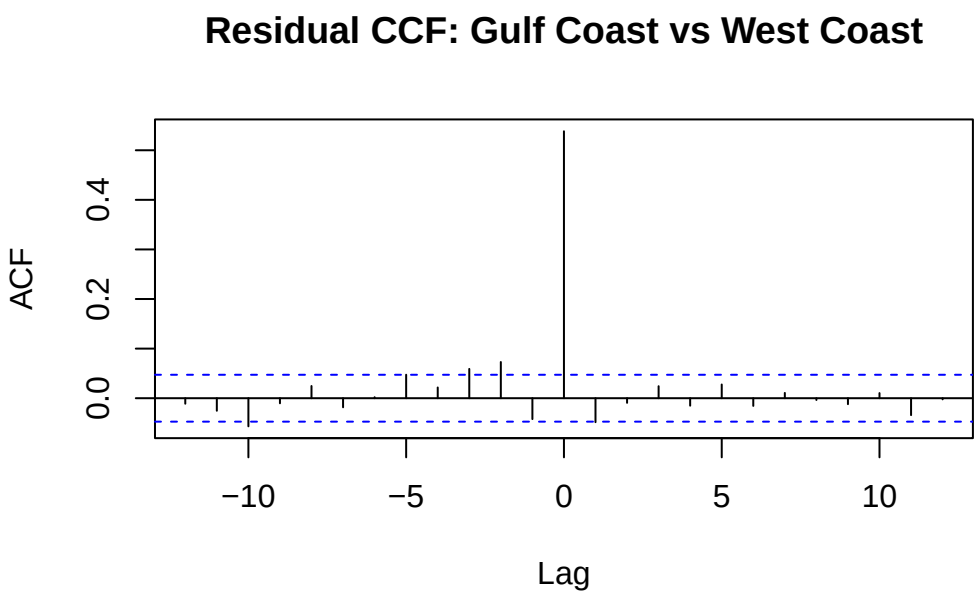

Residual CCF: East Coast vs Midwest
ACF
0.6
0.4
0.2
0.0
−10
−5
0
5
10
Lag


Residual CCF: East Coast vs Gulf Coast
ACF
0.0
0.4
0.8
−10
−5
0
5
10
Lag


Residual CCF: East Coast vs West Coast
ACF
0.0
0.2
0.4
−10
−5
0
5
10
Lag


Residual CCF: Midwest vs Gulf Coast
ACF
0.0
0.2
0.4
0.6
−10
−5
0
5
10
Lag


Residual CCF: Midwest vs West Coast
ACF
0.0
0.2
0.4
−10
−5
0
5
10
Lag

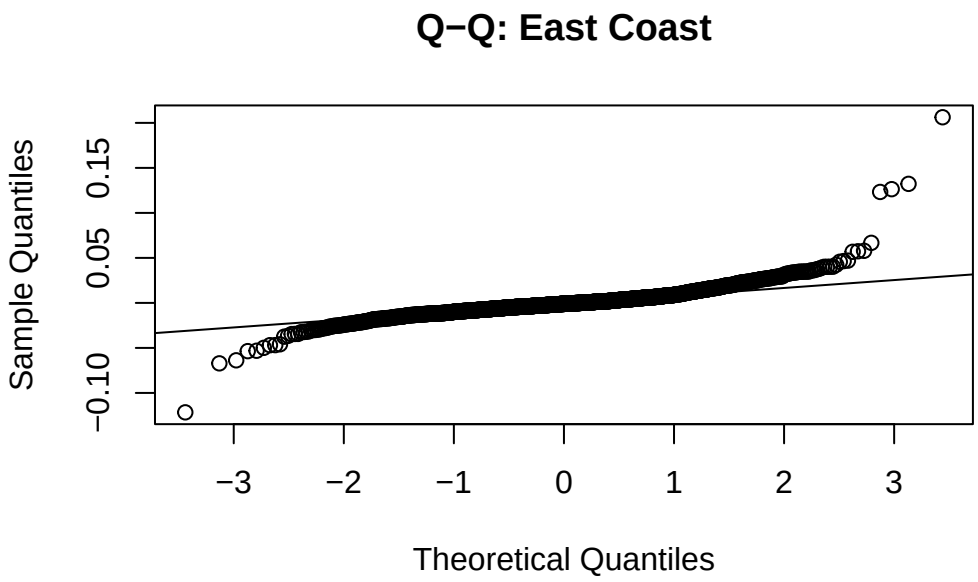


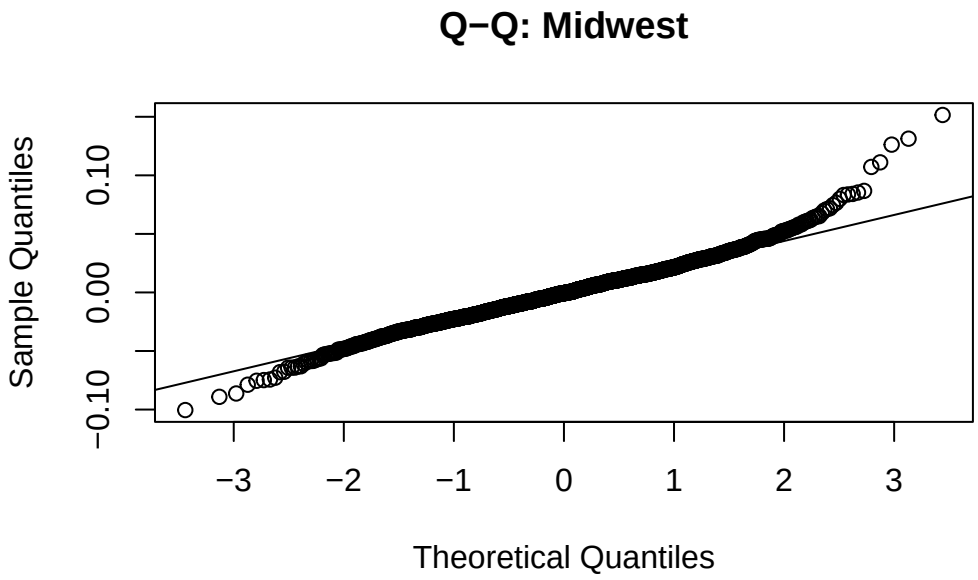

```
par(mfrow = c(1, 1))
```

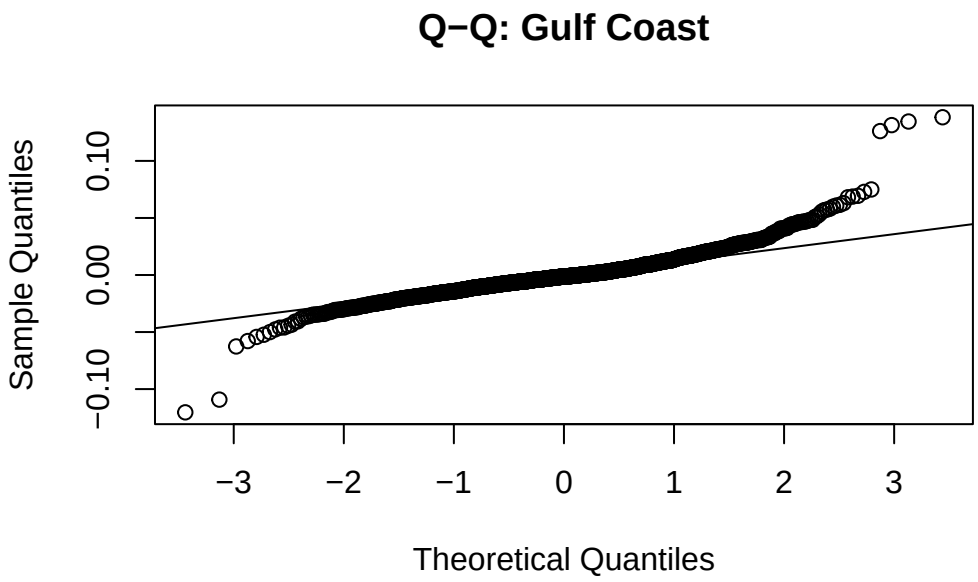
CCF is strange for Rocky Mountain comparison, but only has spikes at lag 0 for all other CCF comparisons.

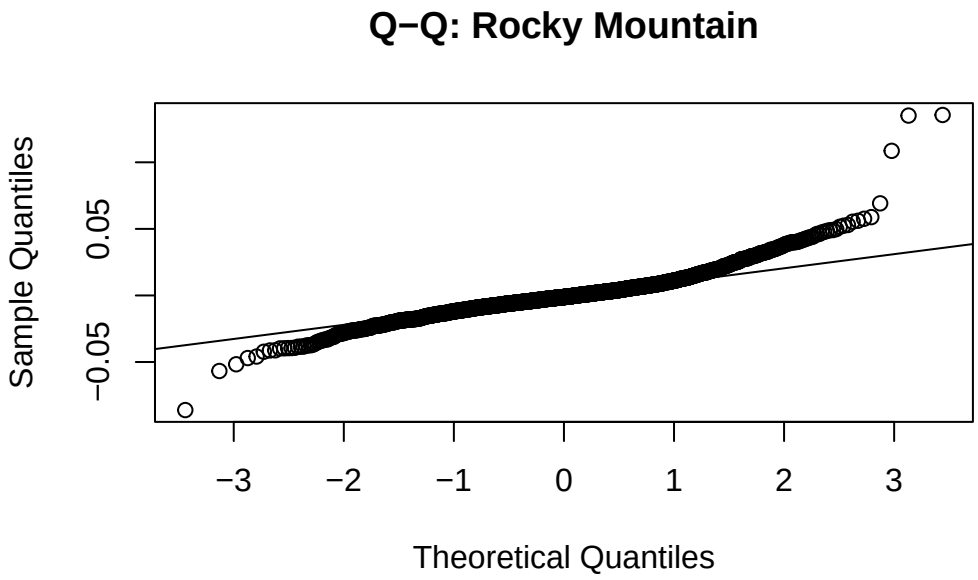
FINISH: why????

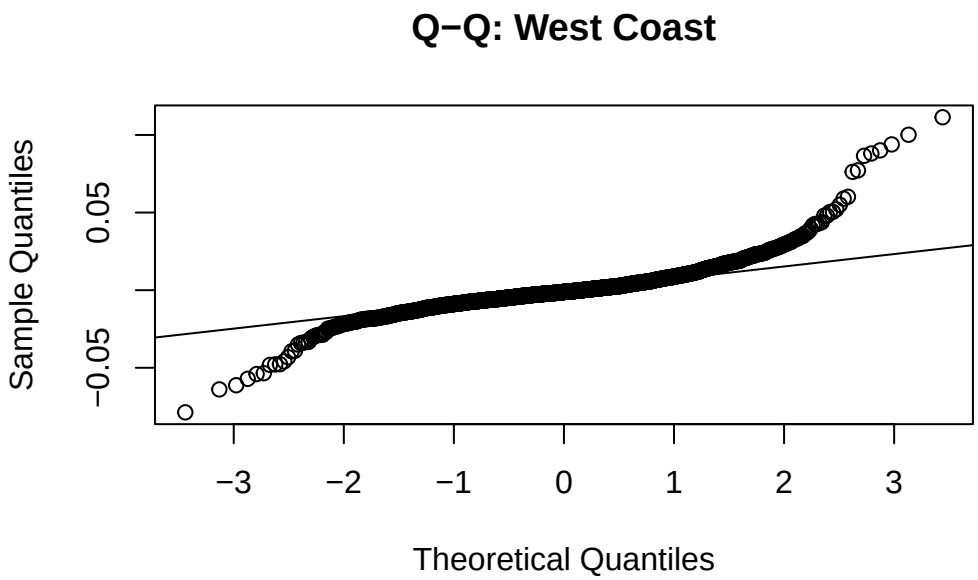
```
# Normal Q-Q plots

for (nm in colnames(res_regions)) {
  qqnorm(res_regions[, nm],
         main = paste("Q-Q:", gsub("_", " ", nm)))
  qqline(res_regions[, nm])
}
```


Q−Q: East Coast
Sample Quantiles
−0.10
0.05
0.15
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Q−Q: Midwest
Sample Quantiles
−0.10
0.00
0.10
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Q−Q: Gulf Coast
Sample Quantiles
−0.10
0.00
0.10
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Q−Q: Rocky Mountain
Sample Quantiles
−0.05
0.05
−3
−2
−1
0
1
2
3
Theoretical Quantiles




```r
par(mfrow = c(1, 1))
```

Heavy tailed expected - maybe look into outliers? deviation from qq-line is likely to due larger shocks.

(v) Granger Causality (pairwise)

Granger causality does not make sense for regional pairs, and any lags between regions can be determined by CCF and VAR coefficients.

(vi) Forecast